Informatica® Cloud Data Governance and Catalog

April 2026

Data Quality for Assets

Informatica Cloud Data Governance and Catalog Data Quality for Assets
April 2026

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Preface

Read *Data Quality for Assets* to learn how to use data quality rules to improve the quality of your organization's data.

CHAPTER 1

Data quality scores

In Data Governance and Catalog, you can view data quality scores for the data in your organization.

If Data Governance and Catalog is integrated with a data quality application, you can define data quality rules in Data Governance and Catalog and associate them with Glossary assets that are linked to technical assets. Although you define the rules in Data Governance and Catalog, the rules are created in the data quality application. Data Governance and Catalog sends the data elements associated with the glossary to the data quality application. After the data quality rules are run on the data elements, Data Governance and Catalog retrieves the data quality scores and displays them in the asset pages. You can also run data quality rules for a catalog source automatically according to the schedule that you define in Metadata Command Center. For more information about running automated data quality rules on Glossary assets, see "Defining data quality rules for business assets" on page 14.

Using the bulk import process to modify data quality scores

You can use the bulk import process to modify the data quality score of a data element that is associated with an asset. You can use this method if you want to update the aggregated data quality score of an asset before its associated data quality rule is next run.

You can also use the bulk import process or run a data quality rule to update the data quality score of a data element that is not linked to any business asset.

CHAPTER 2

Run data quality rules on assets

You can define a data quality rule that you want to run on one or more data elements. The rule is run in the data quality application that is integrated with Data Governance and Catalog. After the rule is run and the results are ready, Data Governance and Catalog retrieves the rule scores and displays the results in graphical formats. You can also specify target and threshold values to categorize rule results as Good, Acceptable and Not Acceptable. For the catalog sources that you configure in Metadata Command Center, you can retrieve the data quality scores for the technical assets associated with the catalog source. You can disable the running of data quality rules for a catalog source in Metadata Command Center.

Prerequisites for running data quality rules

According to the data quality application that you use, you must configure the system and the users so that the data quality rules can be run properly. Before running data quality rules, ensure to enable the **Data Quality** option in Metadata Command Center for each catalog source.

Running data quality rules in Informatica Cloud Data Quality

If you want to define data quality rules in Data Governance and Catalog and run the rules in Informatica Cloud Data Quality, the following prerequisites apply:

- Data Governance and Catalog must be integrated with Informatica Cloud Data Quality.
- You must be able to access Informatica Cloud Data Quality using the same login credentials that you use for Data Governance and Catalog.
- To associate an existing rule in Data Quality with a Glossary asset in Data Governance and Catalog, you must have view permission for the rule specification in Data Quality.
- To create a rule in Data Quality and associate it with a Glossary asset in Data Governance and Catalog, you must have view and execute permissions for rules and profiles in Data Quality.

The organization administrator can assign you the required permissions and privileges in the Administrator service. For more information about assigning asset permissions and feature privileges to users, see the *Organization Administration* help in the Administrator.

Running data quality rules in another data quality application

If you want to define data quality rules in Data Governance and Catalog and run the rules in another data quality application, the following prerequisites apply:

- Data Governance and Catalog must be integrated with the data quality application.
- You must have appropriate permissions and privileges to view, create and execute rules and profiles in the data quality application.

Supported assets for automated data quality rules

You can define and run automated data quality rules for the business and technical asset types.

Business assets

The following list shows the supported business assets:

- Data Set
- Data Quality Rule Template
- Glossary Business Term
- Glossary Domain
- Glossary Subdomain
- Glossary Metric
- Process
- Policy
- System

Technical assets

The following table shows the supported technical assets and the associated data types:

Technical assets	Supported data types
Amazon Athena	The following list shows the supported data types for the asset: - bigint - boolean - char - date - decimal - double - float - int - smallint - string - timestamp - tinyint - varchar
Amazon Redshift	The following list shows the supported data types for the asset: - smallint - integer - bigint - decimal - real - double precision - boolean - char - varchar - date - timestamp - timestamptz - geometry - geography - hllsketch - super - time - timetz - varbyte

Technical assets	Supported data types
Amazon S3	The string data type is supported for delimited CSV technical asset. - The following list shows the supported data types for Avro technical asset: - boolean - int - long - double - string - record - array - The following list shows the supported data types for Parquet technical asset: - boolean - int32 - int64 - int96 - float - double - date - decimal - string - time - map - struct
Google BigQuery	The following list shows the supported data types for the asset: - boolean - date - datetime - float - integer - numeric - string - time - timestamp
Google Cloud Storage	- The following list shows the supported data types for Avro technical asset: - boolean - int - long - double - string - record - array - The following list shows the supported data types for Parquet technical asset: - boolean - int32 - int64 - int96 - float - double - date - decimal - string - time - map - struct

Technical assets	Supported data types
JDBC	The following list shows the supported data types for the asset: - bigint - char(L) - date - decimal (p,s) - float - graphic - integer - numeric (p,s) - smallint - time - timestamp - varchar - vargraphic
Microsoft Azure Data Lake Storage Gen2	- The following list shows the supported data types for Avro technical asset: - boolean - int - long - double - string - record - array - The following list shows the supported data types for Parquet technical asset: - boolean - int32 - int64 - int96 - float - double - date - decimal - string - time - map - struct

Technical assets	Supported data types
Microsoft Azure SQL Server	The following list shows the supported data types for the asset: - bigint - numeric - bit - decimal - int - money - smallint - smallmoney - tinyint - float - real - date - datetime2 - smalldatetime - datetime - time - char - varchar - text - nchar - nvarchar - ntext
Microsoft Azure Synapse	The following list shows the supported data types for the asset: - datetime2 - datetime - date - time - float - real - decimal - numeric - money - smallmoney - bigint - int - smallint - tinyint - bit - nvarchar - nchar - varchar - char
Microsoft Fabric Data Lakehouse	The following list shows the supported data types for the asset: - bigint - bit - char - date - datetime2 - decimal - float - int - real - smallint - varchar

Technical assets	Supported data types
Microsoft Fabric Data Warehouse	The following list shows the supported data types for the asset: - bit - bigint - int - smallint - decimal - numeric - float - real - date - time - datetime2 - char - varchar - uniqueidentifer
Microsoft Fabric OneLake	The following list shows the supported data types for the asset: - **Avro**. boolean, bytes, double, float, int, long, null, json - **Flat file**. nstring, number, string - **JSON**. boolean, double, float, int, long, string - **Parquet**. boolean, byte_array, date, decimal, double, float, int32, int64, int96, string, time, timestamp
Microsoft SQL Server	The following list shows the supported data types for the asset: - bigint - numeric - bit - decimal - int - money - smallint - smallmoney - tinyint - float - real - date - datetime2 - smalldatetime - datetime - time - char - varchar - text - nchar - nvarchar - ntext

Technical assets	Supported data types
Oracle	The following list shows the supported data types for Oracle and Oracle RDS assets: - char - date - number - number (p,s) - timestamp - varchar - varchar2 - nchar - nvarchar2 - double precision / float (126) - timestamp with timezone - timestamp with local timezone - long - interval day to second - interval year to month - binary float - binary double - float - urowid - rowid
SAP ERP	The following list shows the supported data types for SAP ERP assets: - char - clnt - cuky - curr - dats - dec - int1 - int2 - lang - lchr - numc - quan - sstrg - strg - tims - unit

Technical assets	Supported data types
Salesforce	The following list shows the supported data types for the asset: - id - reference - boolean - string - datetime - date - double - url - percent - currency - phone - textarea - email
Snowflake	The following list shows the supported data types for the asset: - number - float/double - varchar - boolean - date - time - timestamp_ltz - timestamp_ntz - timestamp_tz - object - array - variant

Defining data quality rules for business assets

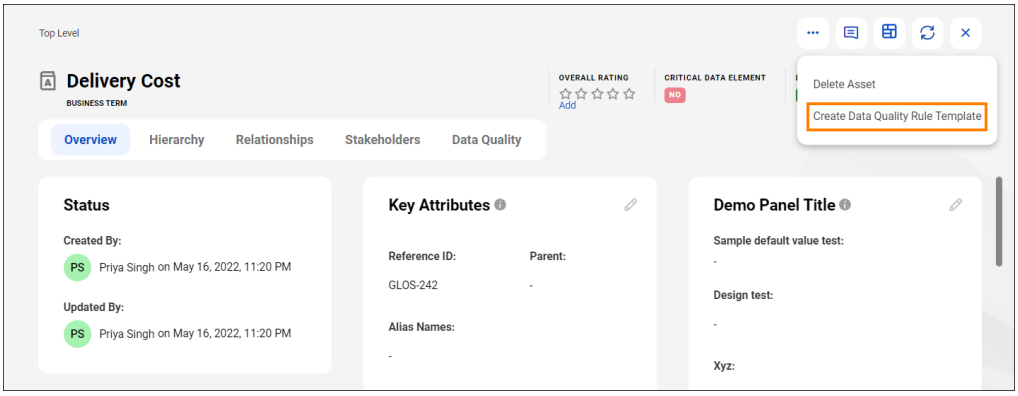
Define a data quality rule to be run on a Glossary asset in Data Governance and Catalog. To define the rule to run on Glossary assets, create a data quality rule template, specify the properties of the rule template, and associate the rule template with the Glossary asset on which you want to run the rule. The rule that you define is created in the data quality application that is integrated with Data Governance and Catalog. The data quality application runs the rule on the technical assets that are linked to the Glossary asset. After the rule is run, Data Governance and Catalog retrieves and displays the data quality scores of the rule.

Before you create a data quality rule template, make sure that the Glossary asset is linked to the technical assets on which you want to run the rules. For more information about linking Glossary assets to technical assets, see the *Asset Management* help.

To manually create a data quality rule template that runs on the data elements linked to the Glossary asset, perform the following steps:

1. Open the Glossary asset on which you want to run the rule, and click the Action menu on the right header of the asset page.

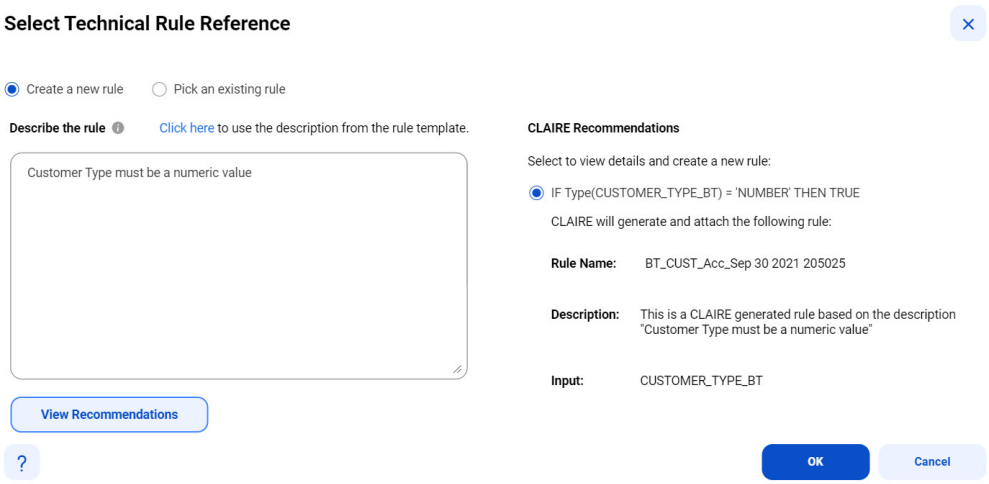
2. From the Action menu, select **Create Data Quality Rule Template**.



3. Alternatively, click **New** > **Business Rules** and select **Data Quality Rule Template**.
4. Enter the following properties of the rule template:

Field	Description
Name	Name of the rule template.
Description	Description of the rule template.
Reference ID	Unique reference identifier for the rule template.
Dimension	Data quality dimension that applies to the data quality rule that is defined by the rule template.
Measuring Method	Method by which the data quality rule for the rule template is evaluated. Select one of the following values: - **Business Extract**. The rule is measured on data that is exported for a particular business case. - **System Function**. The rule is measured by a data quality system. - **Technical Script**. The rule is measured by a script that is manually run by an analyst. - **Informatica Cloud Data Quality**. The rule is measured by Informatica Cloud Data Quality.
Primary Glossary	Glossary asset that is linked to the technical assets on which you want to run the data quality rule. **Note:** Ensure that the first four characters of the primary glossary name don't contain the following special characters: comma (,), hyphen (-), semi-colon (;), backslash (\), single quote ('), double quotes ("), angle brackets (< and >), equal sign (=), parentheses ({ and }), braces ((and)), square brackets ([and]), and period (.)
Secondary Glossary	Additional Glossary assets to which the data quality rule in the rule template applies.

Field	Description
Technical Rule Reference	Create a reference to a rule specification in the data quality application. You can create a new rule specification, or select from an existing rule specification. - To create a new rule specification in the data quality application, click **Create a rule**. Enter a description for the rule using natural language construction, and click **View Recommendations**. CLAIRE® reads the description that you enter and intelligently recommends a rule that it can create in the data quality application. For more information about creating rules that CLAIRE® can interpret, see Guidelines for entering rule descriptions. **Note:** If you want to use the **Create a rule** option, your organization administrator must enable the **Enable CLAIRE Generative AI Services** setting in Administrator. - To select from an existing rule specification in the data quality application, click **Pick an existing rule**. After you select a rule option, click **OK** to go back to the rule template creation page.
Criticality	Criticality of the rule template. Select one of the following values: - High - Medium - Low
Automation	Select the check box to indicate that you want the data quality rule to run automatically on the data elements that are linked to the Primary Glossary. If you select the check box, Data Governance and Catalog automatically runs the rule according to the schedule you specify in the **Frequency** field.
Target	Minimum acceptable data quality value for the asset to be considered "Good." The target value is higher than the threshold value. For example, you can set the threshold value to `50` and the target value to `85`.
Threshold	Minimum acceptable data quality value for the asset to be considered "Acceptable." The target value is higher than the threshold value. For example, you can set the threshold value to `50` and the target value to `85`.
Frequency	Frequency of running the data quality rule that is defined by the rule template. - Select `Daily` to run the rule once in a day. - Select `Weekly` to run the rule once in a week. - Select `Monthly` to run the rule once in a month. The rule is run when you click **Create**.



5. When all the details of the rule template are ready, click **Create**.

When you click **Create**, the data quality application runs the rule on the data elements linked to the Glossary assets. If you selected the **Automation** check box, the data quality application runs the rule according to the schedule you specified in the **Frequency** field. You can now view the data quality scores when you open the rule template, the corresponding rule occurrences, the Glossary asset, and the linked technical assets. If a rule occurrence is created from an automated rule template, the term `Auto-` is prefixed to the name of the rule occurrence.

Defining data quality rules for data elements

Define a data quality rule to be run on a data element that is the part of the metadata extracted by Metadata Command Center. To define the rule, create a data quality rule occurrence, specify the properties of the rule occurrence, and associate the rule occurrence with the data element on which you want to run the rule.

Before you run the data quality rules, enable the **Data Quality** capability for the catalog source in Metadata Command Center.

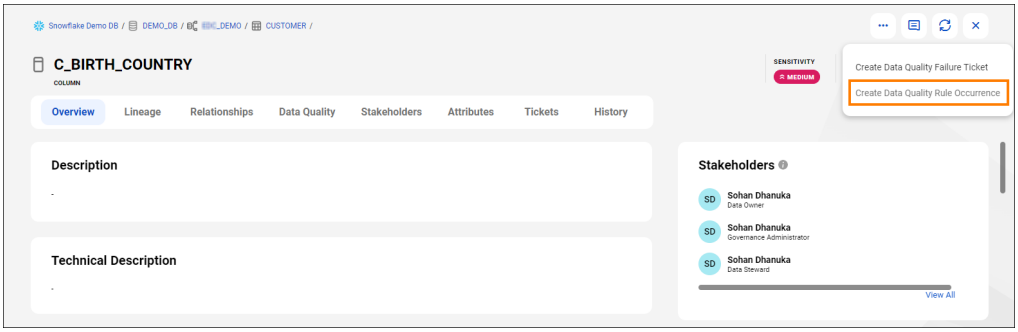
The rule you define is created in the data quality application that is integrated with Data Governance and Catalog. The data quality application runs the rule on the data element. After the rule is run, Data Governance and Catalog retrieves and displays the data quality scores of the data element in the rule occurrence.

Note: The following guidelines apply for associating rule occurrences to data elements:

- The rule occurrences can be manually created, automatically generated, or can be associated with a data quality score card. For automatically generated rule occurrences, you can view the individual data quality scores of the data elements on which the automated data quality rules have been run. You can also manually run a data quality rule to generate manually created rule occurrences.
- When a profile that has a rule associated with it is run in Data Profiling, you can view the data quality score card of the rule occurrence in Data Governance and Catalog. You can't view the total runs of the rule occurrence, and can't preview failed rows immediately after the profile run. The **Total Runs** data appears within one hour after the profile run. Rule occurrences related to the score card are generated in Data Governance and Catalog. You can associate a data element to a rule occurrence that is generated from a score card.

- You can manually associate rule occurrences with data elements. When you purge a catalog source, the system also deletes the associated rule occurrences.

1. Open the data element where you want to run the rule, and click the Action menu in the right header.
2. From the Action menu, select **Create Data Quality Rule Occurrence**.



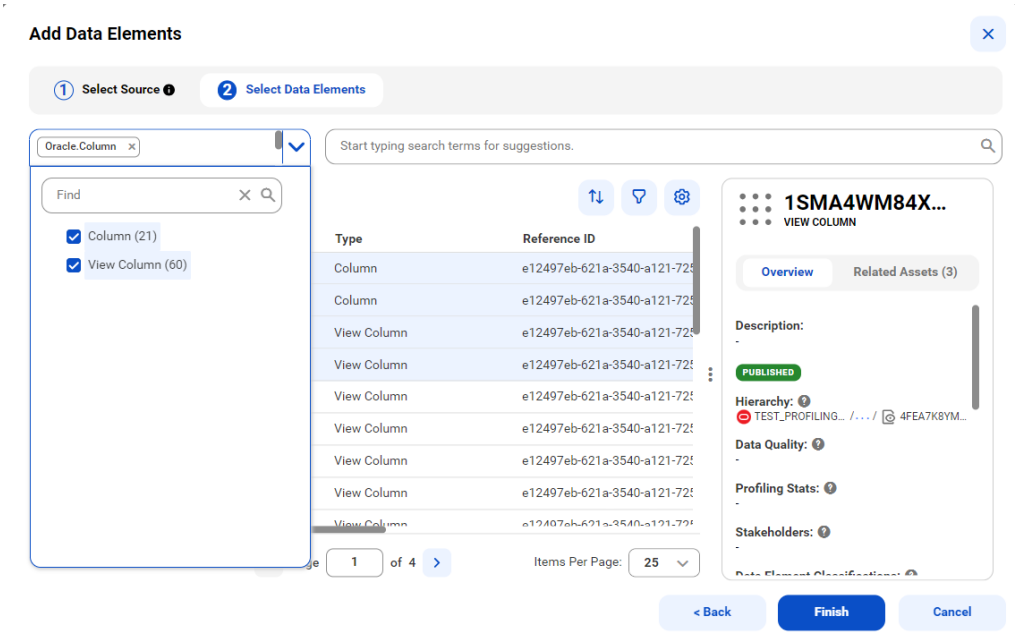
3. Alternatively, click **New** > **Business Rules** and select **Data Quality Rule Occurrence**.
4. Enter the following properties of the rule occurrence:

Field	Description
Rule Template	Data quality rule template that defines the parameters of the rule that you want to run on the data element.
Sync with Rule Template	Syncs the rule occurrence with the specified rule template. When synced, the rule occurrence inherits parameters from the rule template. The rule occurrence score uses the technical rule reference, target, and threshold values from the template. Rule occurrence scores update according to the rule automation schedule in the template. When not synced, the rule occurrence is an independent asset. Template parameters don't affect it.
Name	Name of the rule occurrence.
Description	Description of the rule occurrence.
Reference ID	Unique identifier for the rule occurrence.
Dimension	Data quality dimension for which the data quality rule is run.
Measuring Method	Method by which the data quality rule for the rule occurrence is evaluated. This field can have one of the following values: - **Business Extract**. The rule is measured on data that is exported for a particular business case. - **System Function**. The rule is measured by a data quality system. - **Technical Script**. The rule is measured by a script that is manually run by an analyst. - **Informatica Cloud Data Quality**. The rule is measured by Informatica Cloud Data Quality. **Note:** If you select Informatica Cloud Data Quality as the measuring method, the **Technical Rule Reference** field is mandatory. This field is not mandatory for other measuring methods.

Field	Description
Primary Data Element	Data element where the data quality rule runs. This data element is the input port for the rule in the integrated data quality system. The system always generates the data quality score for the primary data element. To add a primary data element: 1. Select a system asset or catalog source. 2. Select a technical data set or business data set. 3. Select data elements from the data set. If you do not enable the **Data Quality** option in Metadata Command Center for a catalog source, the options in this field are disabled. **Editing the primary data element** Whether you can edit the primary data element depends on how you created the rule occurrence: - Manually created rule occurrences: You can edit the primary data element until you run the rule occurrence for the first time. You can't edit it after the rule runs. - Rule occurrences from data profiling scorecards: You can add the primary data element when you associate the rule occurrence. You can't edit it after you add it. If you don't add a primary data element, no data quality score appears. - Auto-generated rule occurrences: You can't edit the primary data element. You can't edit the primary data element for auto-generated rule occurrences. **Note:** Ensure that the first four characters of the primary data element name don't contain the following special characters: comma (,), hyphen (–), semi-colon (;), backslash (\), single quote ('), double quotes ("), angle brackets (< and >), equal sign (=), parentheses ({ and }), braces ((and)), square brackets ([and]), and period (.)
Secondary Data Element	Secondary input port for the data quality rule that is run on the **Primary Data Element**. You can edit the secondary data element after the rule occurrence is created, but can't edit the element after the rule occurrence is executed.

Field	Description
Technical Rule Reference	Create a reference to a rule specification in the data quality application. You can create a new rule specification, or select from an existing rule specification. - To create a new rule specification in the data quality application, click **Create a rule**. Enter a description for the rule using natural language construction, and click **View Recommendations**. CLAIRE® reads the description that you enter and intelligently recommends a rule that it can create in the data quality application. To create rules that CLAIRE® can interpret, see "Guidelines for entering rule descriptions" on page 25. **Note:** If you want to use the **Create a rule** option, your organization administrator must enable the **Enable CLAIRE Generative AI Services** setting in Administrator. - To select from an existing rule specification in the data quality application, click **Pick an existing rule**. After you select a rule option, click **OK** to go back to the rule occurrence creation page. **Note:** If you select Data Quality as the measuring method, the **Technical Rule Reference** field is mandatory. This field is not mandatory for other measuring methods. When you manually create a rule occurrence, you can specify input parameters from multiple data elements. This evaluates asset quality based on multiple fields. For example, to validate a candidate's hire date, you can use two input parameters: hire date and date of birth. The system uses both parameters to calculate data quality scores. Input parameter mapping is mandatory for a data quality rule occurrence if you are using a multi input parameter rule from the integrated data quality system. Data quality scores are always reported on the primary data element. Map each rule input parameter to a unique data element. Make sure that at least one rule input parameter is mapped to a primary data element.
Criticality	Criticality of the rule occurrence. The value can be `High`, `Medium`, or `Low`.
Target	Minimum acceptable data quality value for the asset to be considered "Good." The target value is higher than the threshold value. For example, you can set the threshold value to `50` and the target value to `85`.
Threshold	Minimum acceptable data quality value for the asset to be considered "Acceptable." The target value is higher than the threshold value. For example, you can set the threshold value to `50` and the target value to `85`.
Frequency	Frequency of running the data quality rule that is defined by the rule template. - Select `Daily` to run the rule once in a day. - Select `Weekly` to run the rule once in a week. - Select `Monthly` to run the rule once in a month. The rule is run when you click **Create**.

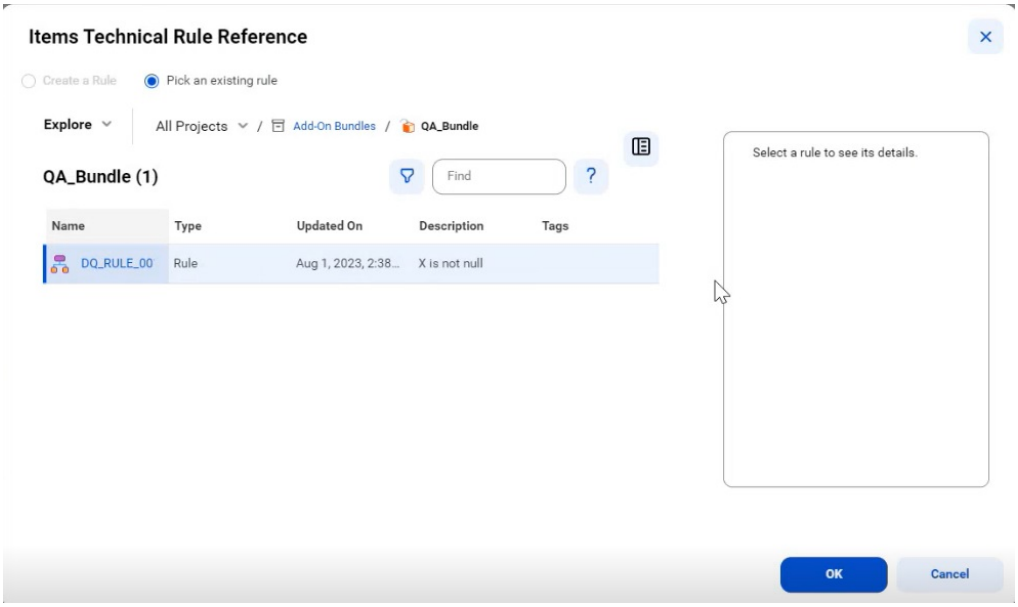
The following image shows the dialog box to select the primary data element:



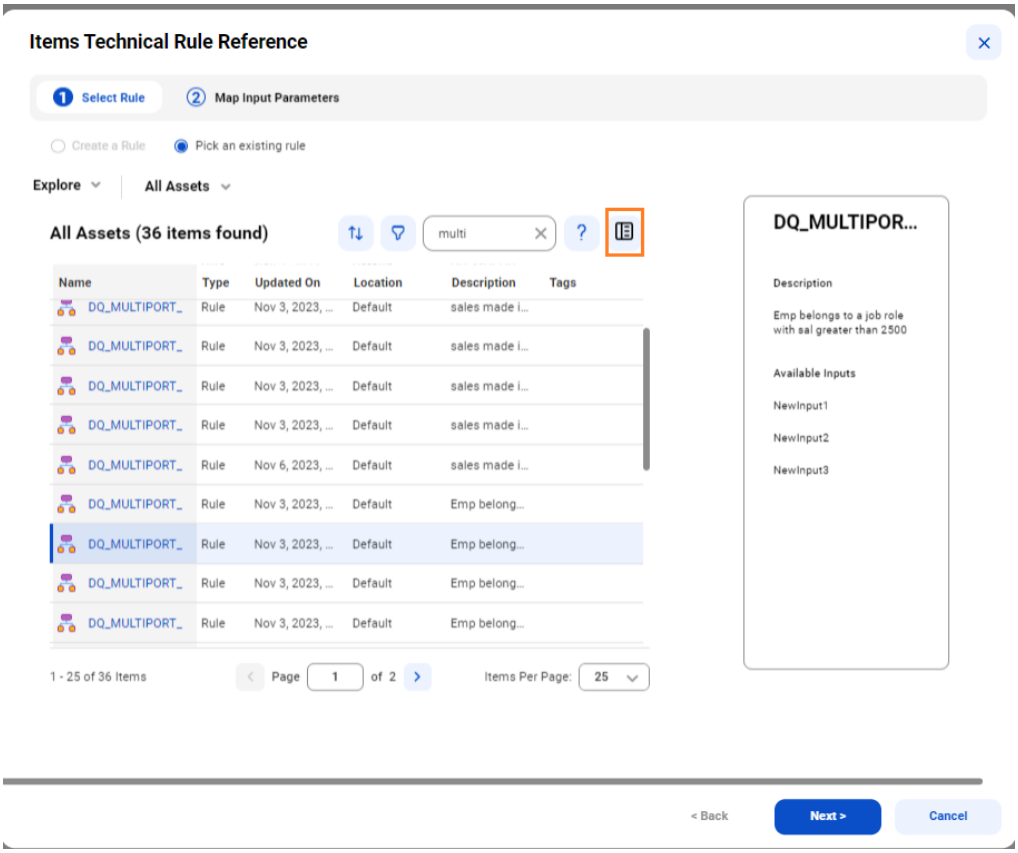
The following example image shows a rule description and its CLAIRE® recommendation:



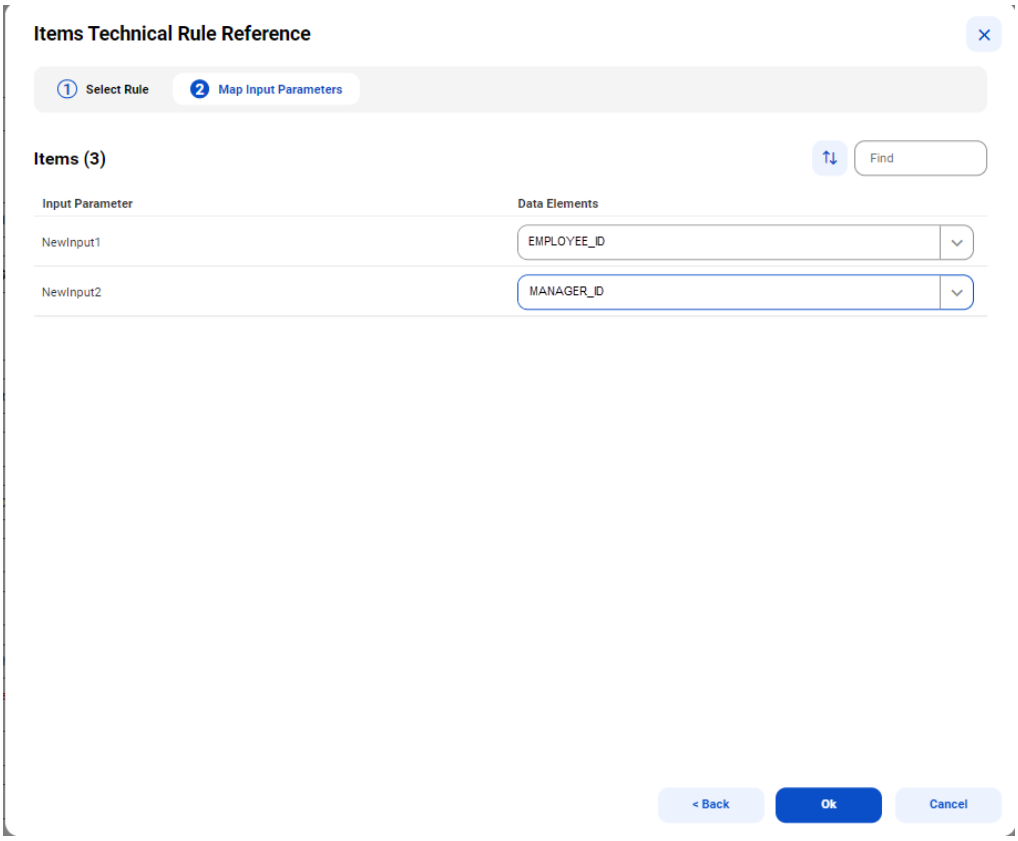
The following image shows the addition of rule reference for a single input parameter from an existing rule specification:



The following image shows the addition of a rule with multiple input parameters and the **Show Preview** icon to view the rule:

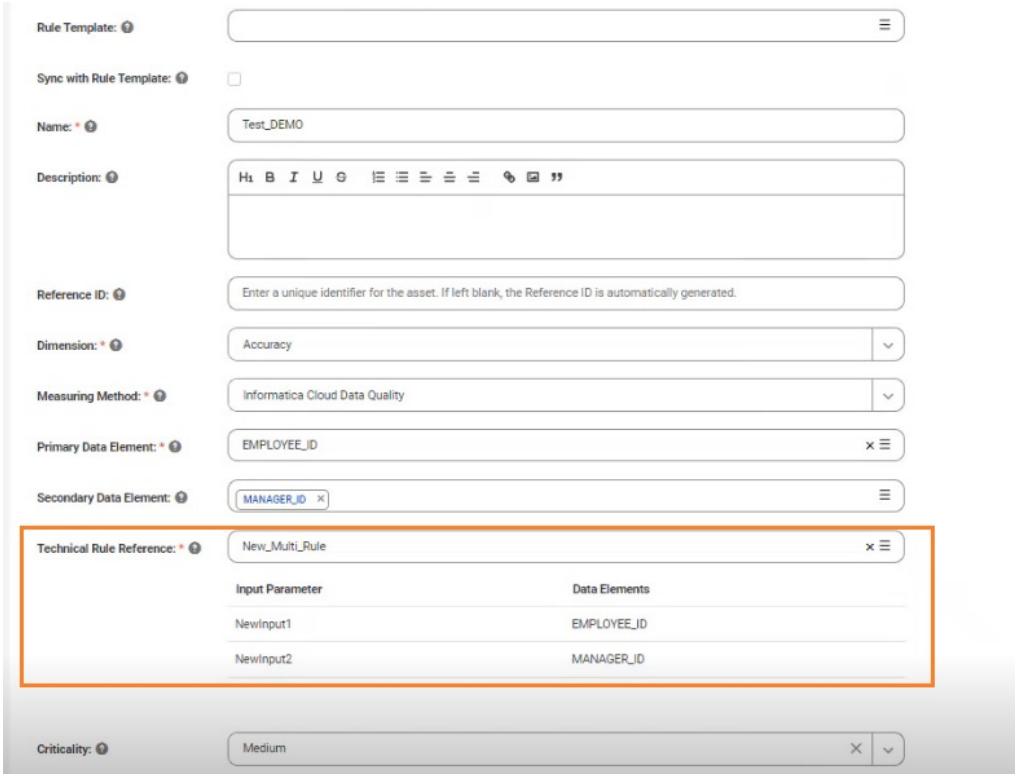


The following image shows how to map multiple input parameters to data elements:



The following image shows the input parameters mapped to the primary and secondary data elements:

Figure 1.

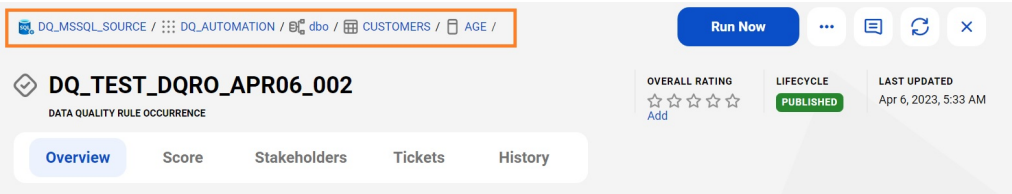


5. When all the details of the rule occurrence are ready, click **Create**.

When you click **Create**, the data quality application runs the rule on the data element. If you selected a schedule in the **Frequency** field, the data quality application runs the rule on the data element according to the rule schedule you specified in the **Frequency** field. When the data quality score is ready, Data Governance and Catalog displays the score in the rule occurrence.

Technical hierarchy of a rule occurrence

For an automatically generated rule occurrence, the catalog source is the highest unit and the data elements are the smallest units in the technical hierarchy.



Note: If you bulk import the assets, the rule occurrences do not have the mentioned technical hierarchy. As a result, the rule occurrences are not removed when you purge a catalog source.

For more information, see the following topics:

- "Run an automated data quality rule" on page 28
- "Run a data quality rule manually" on page 33
- Chapter 3, "View data quality scores in assets" on page 35

Guidelines for entering rule descriptions

When you provide a description for a data quality rule template, CLAIRE® uses Natural Language Processing (NLP) technology to analyze the text and suggest a suitable rule to create in the data quality application. You can review the recommended rule and decide whether the rule meets your requirement. If you agree to the recommendation, Data Governance and Catalog creates the rule in the data quality application and associates the rule with the rule template.

Consider the following guidelines when you enter a rule description so that CLAIRE® can recommend appropriate rules:

- The description can be in any language.
- The description must be within 200 characters and 30 words.
- The description must be in one sentence, and can contain letters, numbers, spaces, and special characters.
- The description can contain UTF-8 characters, spaces, and the following symbols: comma (`,`), hyphen (`-`), semi-colon (`;`), backslash (`\`), single quote (`'`), double quotes (`"`), angle brackets (`<` and `>`), equal sign (`=`), parentheses (`{` and `}`), braces (`(` and `)`), square brackets (`[` and `]`), and period (`.`)
- Avoid spelling errors while creating a rule.
- If the name of the asset on which you want to run a data quality rule contains an unsupported character, omit the character in the rule description. For example, if a business term is called `Customer_Name`, remove the underscore and enter `CustomerName` in the rule description.
- Use simple rules in an NLP rule description. If the rule description contains two or more conditions, CLAIRE® generates the rule that is only for the first condition and ignores the rest. For example, if a rule description contains `Column should be not null and greater than 10`, CLAIRE® generates a rule only for `not null` and ignores `greater than 10`.

NLP texts to consider

You can use the following types of NLP rule descriptions. If the text you enter is insufficient or unclear, CLAIRE® might recommend that you rephrase the rule description.

Empty Value Rules

The following examples show descriptions of null value rules that CLAIRE® NLP technology can read:

- `Sum may be NULL`
- `keep the section field empty`
- `Box value should be nullified`
- `Title should be Nothing`
- `Blank values are equal to null`

Non-Empty Value Rules

The following examples show descriptions of rules that are not null values that CLAIRE® NLP technology can read:

- `Age is always NOT NULL`
- `Name field cannot be blank or empty`
- `RFID values is not Null`

- `Radius must not be blank or empty`
- `doc number is not NULL`

Comparison Rules

The following examples show descriptions of comparison rules that CLAIRE® NLP technology can read:

- `Down payment number is greater or equal to thousand`
- `Down payment is going to either be 1000 or be above 1000`
- `Value cannot be less than 1,000`
- `Intensity is lesser than 1,000 lumes`
- `Bill number is larger or equal to zero`

Range Rules

The following examples show descriptions of range rules that CLAIRE® NLP technology can read:

- `Diameter exceeds 0 and doesn't exceed 5`
- `Diameter is bigger than zero and also Diameter is smaller than five`
- `Diameter is between one and five, but it cannot be one or five`
- `Diameter is at least 1 or higher, but not as high as 10`
- `Diameter can be equal to 1, or be between 1 and 10`

Length Rules

The following examples show descriptions of length rules that CLAIRE® NLP technology can read:

- `Zip code must not be longer than 6 characters`
- `Description should not be greater than 180 letters`
- `house rent should not exceed beyond 3000`
- `Only one sentence`
- `Oil Rig ID is 9 characters long`

List Rules

To define a list, enter markups or delimiters, such as curly braces (`{}`) and semicolon (`;`). To facilitate naturally written sentences, you can include conjunctions in the delimiters. Enclose each item of a list within double quotes (`""`).

The following examples show descriptions of list rules that CLAIRE® NLP technology can read:

- `The value of Index is "0", "1" or "2"`
- `Index shall be equivalent to "yes", "no" or "NA"`
- `Index will be one of these: "yes", "no", "NA"`
- `Index should have values within the limit of "yes", "no" and "NA"`
- `"Yes", "no" or "na" can only be the values that Index can hold`
- `The values "NA", "Nil", or "None" shouldn't be present in an email ID`
- `Employee ID should not have values within the limit of "0", "1", or "2"`

Note: Use double quotes to indicate lists. If at least one double quote is found in the sentence, CLAIRE® NLP technology attempts to read the description as a list rule.

Date Rules

The following examples show descriptions of date value rules that CLAIRE® NLP technology can read:

- `Expiration Date must be a date`
- `Birthday must be a date`
- `DOB can have values of type date only`
- `Joining Date should contain the value of date`
- `SSN is not of type date`

Number Rules

The following examples show descriptions of number value rules that CLAIRE® NLP technology can read:

- `Last name isn't a number`
- `Number of Seats must be a number`
- `Salary is numeric`
- `Phone Number is of type Numeric`

Note: A rule description should contain only a single condition. CLAIRE® generates the rule only for the first condition when a rule description contains multiple conditions joined by "and" or "or", such as length and date rules, or null and length rules. It ignores all the remaining conditions. The generated rule appears valid without errors, but is incomplete.

NLP texts to avoid

Avoid the following types of NLP rule descriptions. If the text you enter is insufficient or unclear, CLAIRE® might recommend that you rephrase the rule description.

Double Negation Sentences

The following examples show descriptions of double negation sentences that CLAIRE® NLP technology cannot read:

- `It is not defined city name to be not null`
- `Distance is not equal to or lesser than NOT NULL`
- `Date of birth cannot be not null`
- `It is not acceptable for Phone Number to be not NULL`

Complex Sentences

The following examples show descriptions of complex sentences that CLAIRE® NLP technology cannot read:

- `It’s not acceptable for date to equal NULL`
- `Nothing should be populated in name field`
- `Joining Date should not contain the value of date but contains numbers`
- `Product is not equal to 2 or not larger than 1`

Grammatically Incorrect Sentences and Incorrect Spellings

The following examples show descriptions of grammatically incorrect sentences and incorrect spellings that CLAIRE® NLP technology cannot read:

- `First name should be $NULL`

- `Serial cannot contain numerics`
- `Last_name is not a number and alphasnumeric`
- `Product is nt equal to or nt larger then 1`

Run an automated data quality rule

When you define an automated data quality rule, Data Governance and Catalog runs the rules automatically according to the schedule you have specified in the **Frequency** field of the rule template or rule occurrence. You don't need to do anything. You can open the rule occurrence to see the individual data quality scores of the date elements on which the rules have been run.

Consider the following scenarios while you run a data quality rule automation process:

1. The **Data Quality** and **Data Quality Rule Automation** options are enabled in Metadata Command Center for a catalog source. The data quality rule occurrences are created automatically for the catalog source and you can manually run them on the data elements associated with the catalog source.
2. The **Data Quality** option is enabled and **Data Quality Rule Automation** option is disabled in Metadata Command Center for a catalog source. Automatic rule occurrences are not created. However, you can still run the existing data quality rule occurrences on the data elements associated with the catalog source. You can manually create and run the data quality rule occurrences for the catalog source.

Note:

- Data quality automation is not supported on the external tables for Amazon Athena catalog source.
- Before you run an automated data quality rule, ensure to enable the **Data Quality** and the **Data Quality Rule Automation** options in Metadata Command Center for a catalog source.

Automated rules for governance assets

To run automated rules on governance assets, make sure that you have done the following tasks when you define the rule:

- Select the **Automation** check box.
- Specify the rule schedule in the **Frequency** field.

When you click **Create**, Data Governance and Catalog runs the rule once immediately, and then runs all subsequent rules according to your specified schedule.

For more information on defining rules for governance assets, see "Defining data quality rules for business assets" on page 14.

Automated rules for data elements

To run automated rules on data elements, make sure that you select a schedule in the **Frequency** field when you define the rule.

When you click **Create**, Data Governance and Catalog runs the rule once immediately, and then runs all subsequent rules according to your specified schedule.

For more information on defining rules for data elements, see "Defining data quality rules for data elements" on page 17.

Data quality rule automation process

Before you run a data quality rule automation process, ensure to enable the **Data Quality** and **Data Quality Rule Automation** options in Metadata Command Center for a catalog source. To define an automated data quality rule that you want to run on data elements, ensure that the assets are correctly associated with each other in Data Governance and Catalog.

Before you define an automated rule

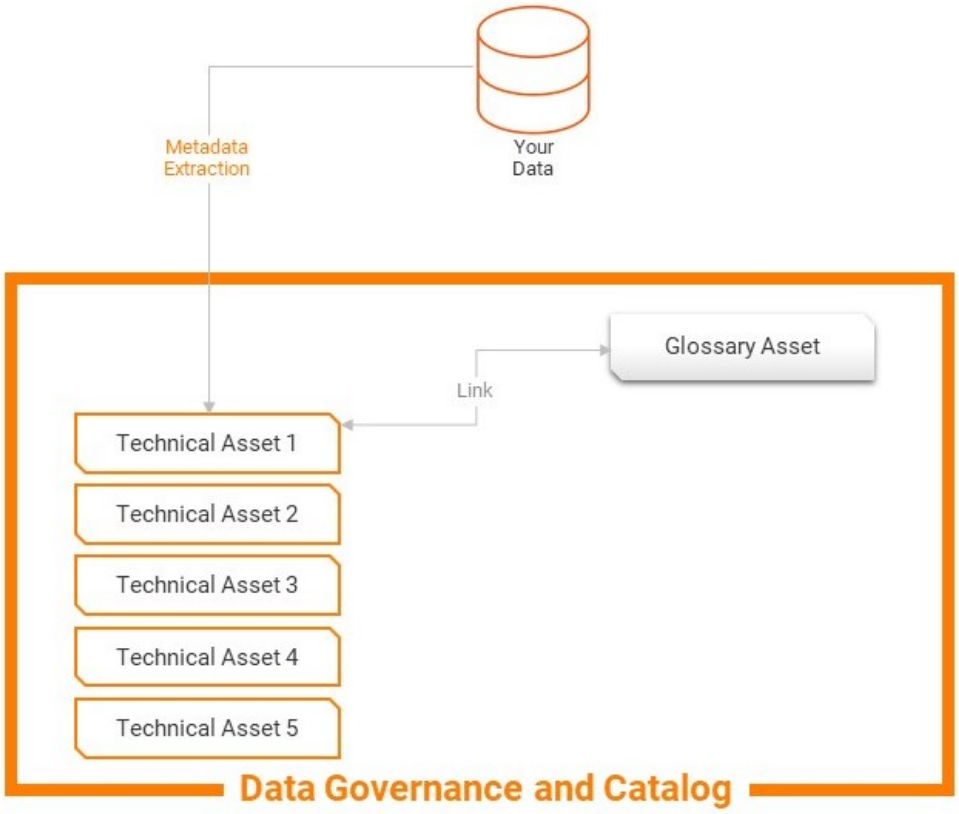
Before you define an automated data quality rule, your must link the technical assets that represent the data elements to the corresponding Glossary business assets.

What is a technical asset? When you define a data quality rule, the rule is run on the data that is located in the source systems in your organization. Metadata Command Center extracts the metadata from the source systems and sends the metadata to Data Governance and Catalog. This metadata appears as technical assets in Data Governance and Catalog. Therefore, technical assets in Data Governance and Catalog are metadata representations of the data in your organization. When you open a technical asset, you can see the constituent data elements. Technical assets also display catalog metadata such as profiling and classification information for the data. For more information about technical assets, see the *Understanding Technical Assets* help.

What is a business asset? Business assets are semantic representations of technical assets. You can enrich the metadata of technical assets by adding governance metadata to the business assets. For example, in a business asset, you can specify a meaningful name, enter a description, and denote the stakeholders of the data assets. For more information about business assets, see the *Understanding business assets* help.

How are technical and business assets related? Technical assets appear in Data Governance and Catalog because Metadata Command Center periodically extracts and sends the metadata. As a user, you create business assets in Data Governance and Catalog. The business assets that you create are independent assets. This is why you must link business assets with their corresponding technical assets. When the linking is complete, a business asset and a technical asset eventually represents the same data set that resides in your organization. You can define a data quality rule only for a business asset of the Glossary type. The rule, however, runs on the data elements that are represented by the Glossary asset or the corresponding technical asset.

The following diagram depicts the link that you must create between a technical asset and a Glossary asset:



For example, you might have five tables in your organization that contain employee names and personal information. These five tables appear as five technical assets in Data Governance and Catalog. To govern these five tables, you can create a single business asset called "Employee Record" of the Glossary Business Term type, and link the business term to the five technical assets. If you now run a mobile number check rule for the business term, the rule is run on all five tables.

For more information about linking a technical asset to a Glossary asset, see the *Asset Management* help.

Defining an automated rule

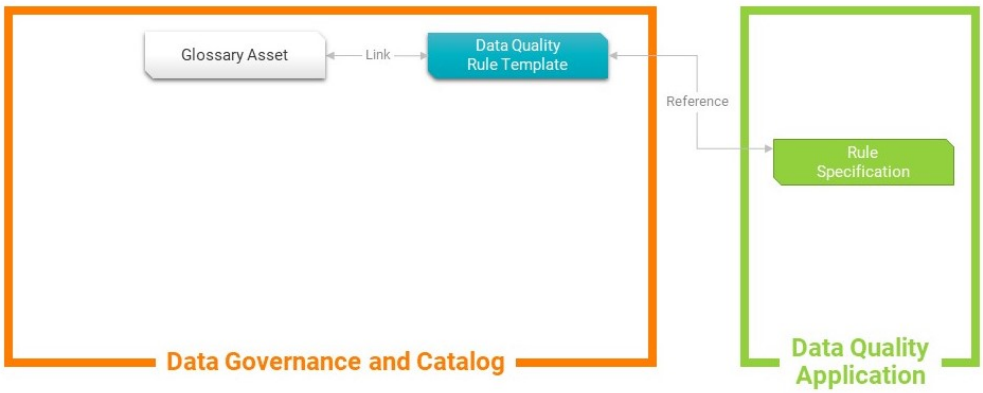
Define an automated data quality rule as a data quality rule template in Data Governance and Catalog, and associate a Glossary asset to the rule template.

What is a data quality rule template? A data quality rule template is the definition for a rule. It contains a textual description of the rule and other essential rule parameters such as dimension, threshold value, target value, and criticality. In the rule template, you must provide a reference to a rule specification in your data quality application. You can either provide a reference to an existing rule specification in your data quality application, or you can create a new rule specification in your data quality application from the rule template page in Data Governance and Catalog. When you provide this reference, a link between the rule template in Data Governance and Catalog and the rule specification in the data quality application is created. When this link is created, the rule template becomes the business representation of the data quality rule in your data quality application. For more information about the fields of a rule template asset, see the *Data Quality Rule Template* topic in the *Understanding Business Assets* help.

How is a rule template and rule specification related? When you create a data quality rule template, you must specify a Glossary asset that you want to associate with the rule template. This association specifies the data elements on which the data quality rule must run. The parameters that you define in the rule template

are applicable to the data quality rule that is run on the data elements that are represented by the Glossary asset. If you do not associate the rule template with a Glossary asset, the rule template remains an independent asset, and there are no data elements on which the specified rule must run.

The following diagram depicts the association that you must create between a data quality rule template and a Glossary asset:



For example, you can create a data quality rule template called "Mobile Number Check" in Data Governance and Catalog and link the rule template to the `Default/mob_phno_check` rule specification in your data quality application.

For more information about creating a data quality rule template, see "Defining data quality rules for business assets" on page 14.

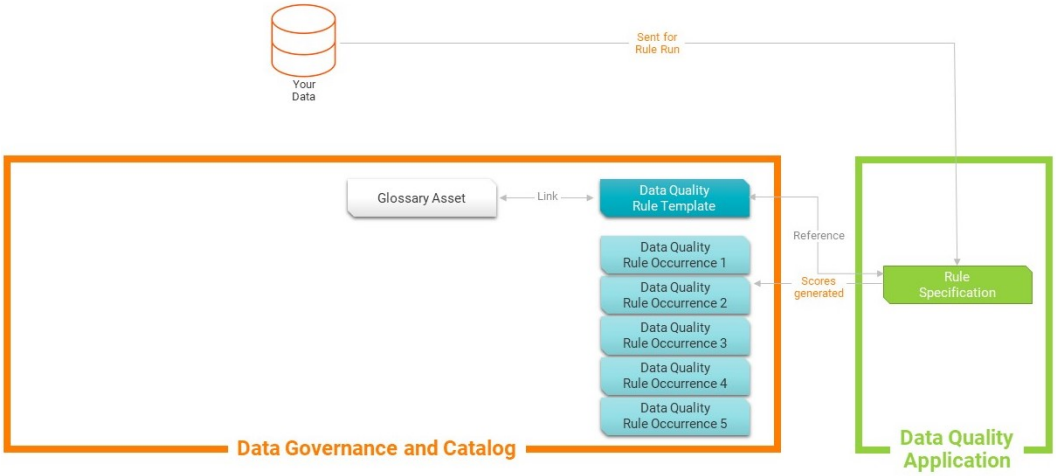
After you define an automated rule

When you create a rule template, the data quality rule is automatically run on the data elements represented by the Glossary asset, and Data Governance and Catalog displays the rule scores as rule occurrences.

What happens when you create a rule template? After you create a rule template, Data Governance and Catalog sends the data elements of the Glossary asset to the data quality application. The rule specification that you specified in the rule template is run on the data elements, and the data quality application generates a rule score for each data element. The data quality application sends the rule scores to Data Governance and Catalog, where it appears as data quality rule occurrences.

What is a data quality rule occurrence? A data quality rule occurrence is the representation of the data quality score of a single data element. While the data quality rule template is the definition of a rule, a data quality rule occurrence is the instance of the rule that is run on a single data element. For more information about the fields of a rule occurrence asset, see the *Data Quality Rule Occurrence* topic in the *Understanding Business Assets* help.

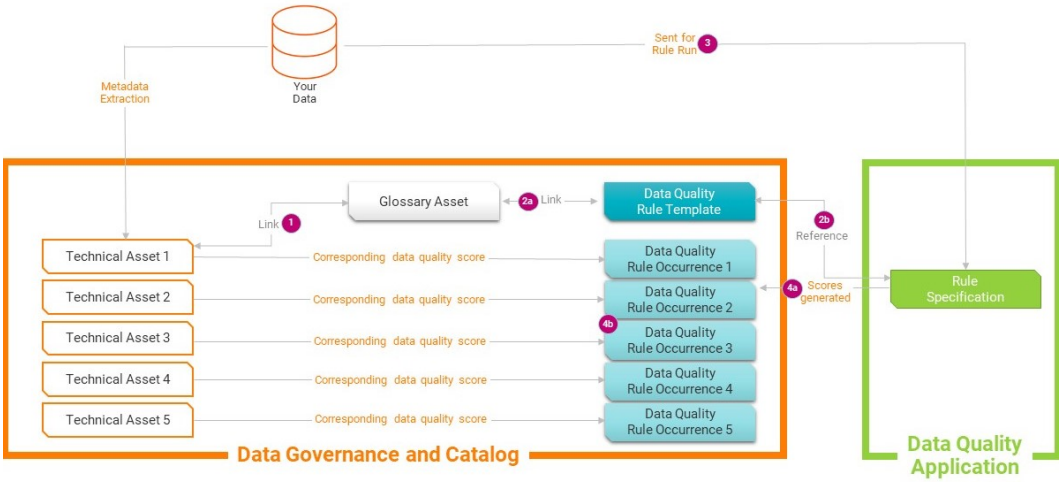
The following diagram depicts the rule occurrences that are created when a data quality rule is run on data elements:



For more information about viewing data quality scores, see Chapter 3, “View data quality scores in assets” on page 35.

Putting it all together

The following diagram depicts the entire process of data quality rule automation:



1. Link the technical assets that represent the data elements to the corresponding Glossary asset.
2. Create a data quality rule template.
    a. In the rule template, specify the Glossary asset that is linked to the technical assets on which you want to run the data quality rule.
    b. Create a reference between the rule template in Data Governance and Catalog and a rule specification in your data quality application.
3. When you save the data quality rule template, Data Governance and Catalog sends the data elements to the data quality application.
4. The data quality application runs the rule on the data elements and generates data quality scores.
    a. The data quality scores appear in Data Governance and Catalog as rule occurrences.
    b. Each rule occurrence corresponds to a data element on which the data quality rule was run.

Run a data quality rule manually

Apart from the automated data quality rule that you might have configured, you can manually run a data quality rule on governance assets any time.

You might want to manually run a data quality rule for a business asset in the following situations:

- When you defined the rule template, you did not select the **Automation** check box.
- When you defined the rule template, you selected the **Automation** check box and specified the schedule in the **Frequency** field.
- You want to run the rule at a time other than the schedule you specified in the **Frequency** field of the rule template. For example, if you specified a weekly schedule, you want to run the rule before one week has elapsed.

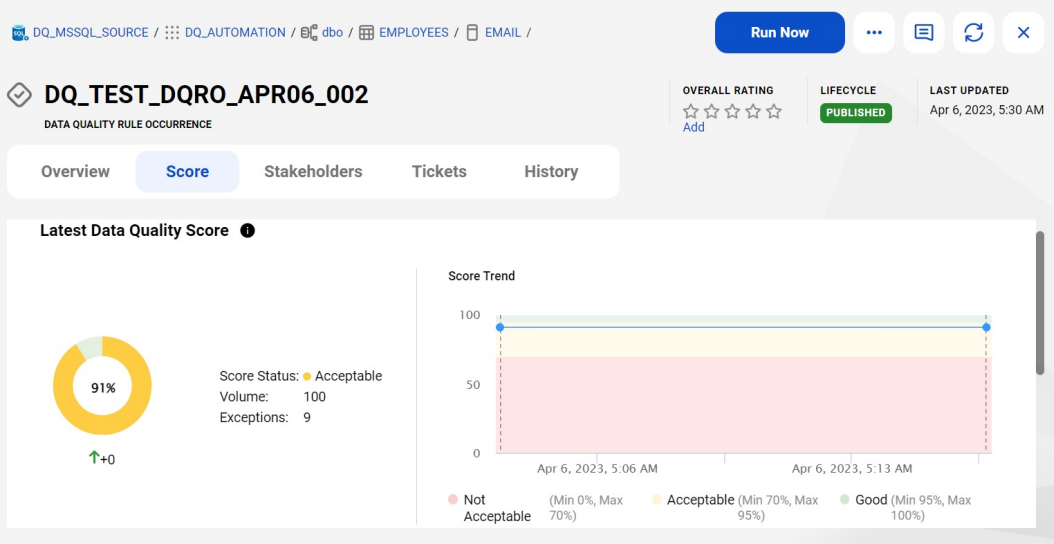
Before you run a data quality rule manually, the following prerequisites apply:

- The **Data Quality** option for the catalog source is enabled in Metadata Command Center.

- Your organization administrator has granted you the **Execute** permission on data quality rule occurrence assets through access policies in Metadata Command Center.
- You're a stakeholder of the data quality rule occurrence.
- The data quality rule occurrence is in the **Published** lifecycle status.

To run a data quality rule manually, go to the **Score** tab of the rule occurrence page and click the **Run Now** option on the top right corner of the page. This triggers a job that retrieves the latest data quality score for the rule occurrence. When the job is triggered successfully, you can monitor the job in Metadata Command Center.

The following image shows the **Score** tab of a data quality rule occurrence:



For information about defining rules for business assets, see "Defining data quality rules for business assets" on page 14.

CHAPTER 3

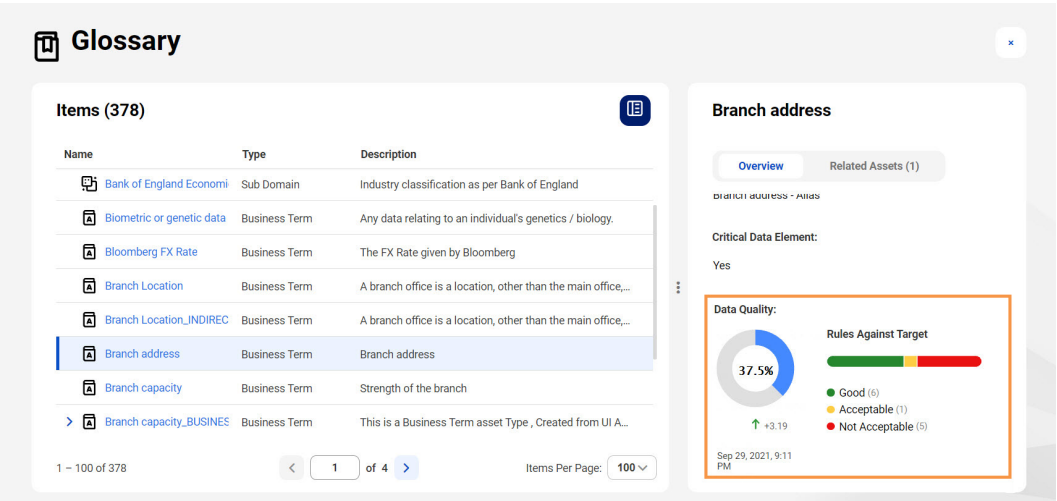
View data quality scores in assets

You can view different types of data quality scores for assets in Data Governance and Catalog. The data quality scores are not immediately propagated during a relationship change between the assets. You cannot view the data quality scores for business assets with `Obsolete` lifecycle status.

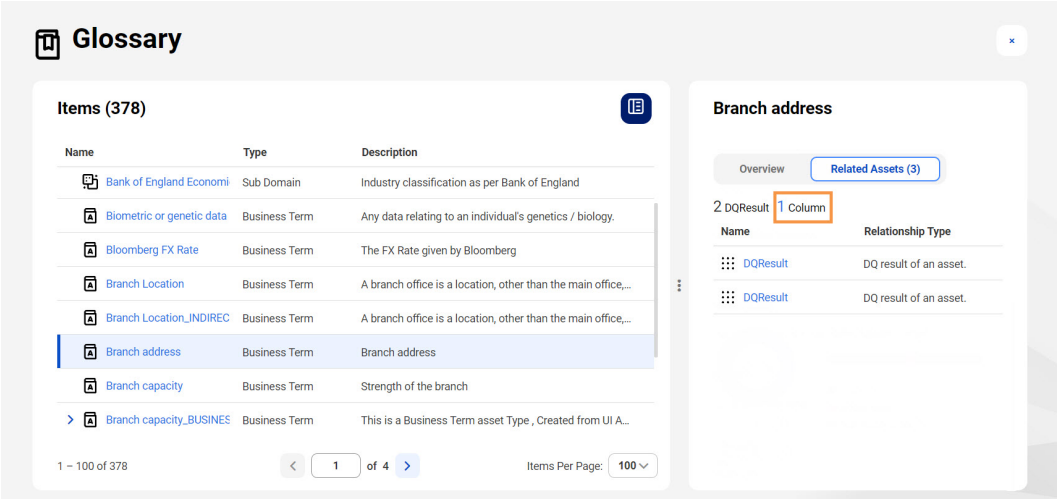
Preview pane of an asset

When you select an asset row in a search result, the preview pane shows a summary of the data quality result for the asset. The result is the composite score of all data elements that comprise the asset. The summary can help you quickly determine whether you need to open the asset in a separate page to view its details.

Search for an asset, select the asset row in the search result, and go to the **Overview** tab of the preview pane. The following example image shows the data quality score for the 'Branch address' business term:



If you open the preview pane of a business asset, go to the **Related Assets** tab to see the linked technical assets for which the data quality scores are displayed. The following image shows the **Related Assets** tab on the preview pane of the 'Branch address' business term:

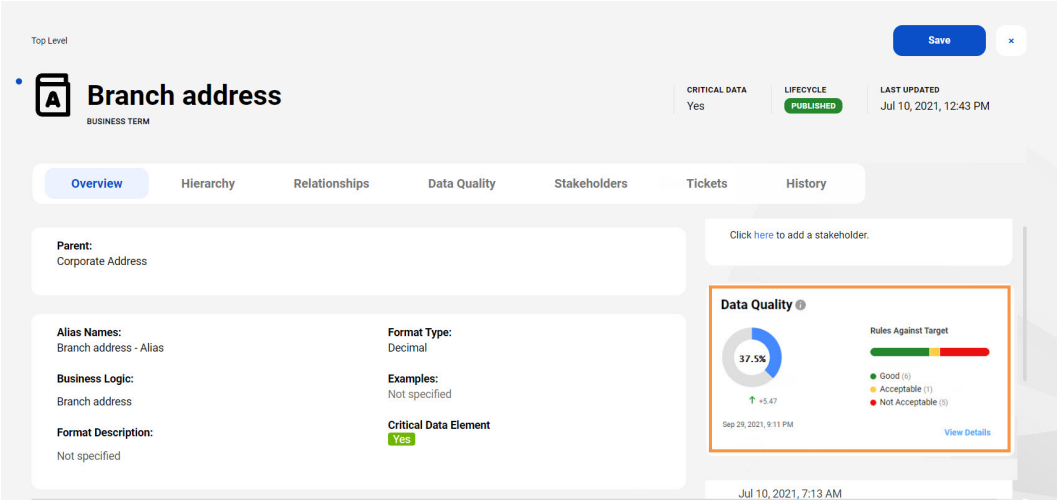


Overview tab of an asset

When you open an asset in a new page, the **Data Quality** panel in the **Overview** tab shows a snapshot of the data quality result of the asset.

The result is the composite score of all data elements that comprise the asset. If the asset that you open is of the Glossary type, the score is the data quality value of the technical assets that are linked to the Glossary asset. If the asset that you open is a technical asset, the score is the data quality value of the data elements that are contained in the technical asset.

The following image shows the data quality score on the **Overview** tab of a business term:

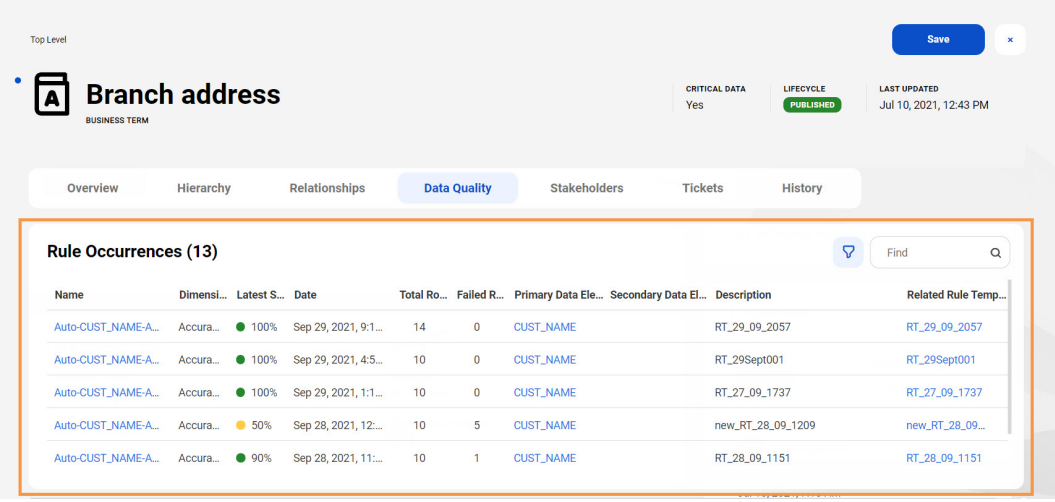


Data Quality tab of an asset

When you open an asset, the **Data Quality** tab shows all rule occurrences that are manually created and automatically generated from the rule template.

A rule occurrence represents the instance of a data quality rule that is run on a single data element. For example, if a table has 10 rows, the **Data Quality** tab will show 10 rule occurrences. The rule occurrences are the individual data quality scores that constitute the composite score that appears on the **Overview** tab of the asset. If the asset that you open is of the Glossary type, the score in the **Data Quality** tab is the data quality value of the technical assets that are linked to the Glossary asset. If the asset that you open is a technical asset, the score in the **Data Quality** tab is the data quality value of the data elements that are contained in the technical asset.

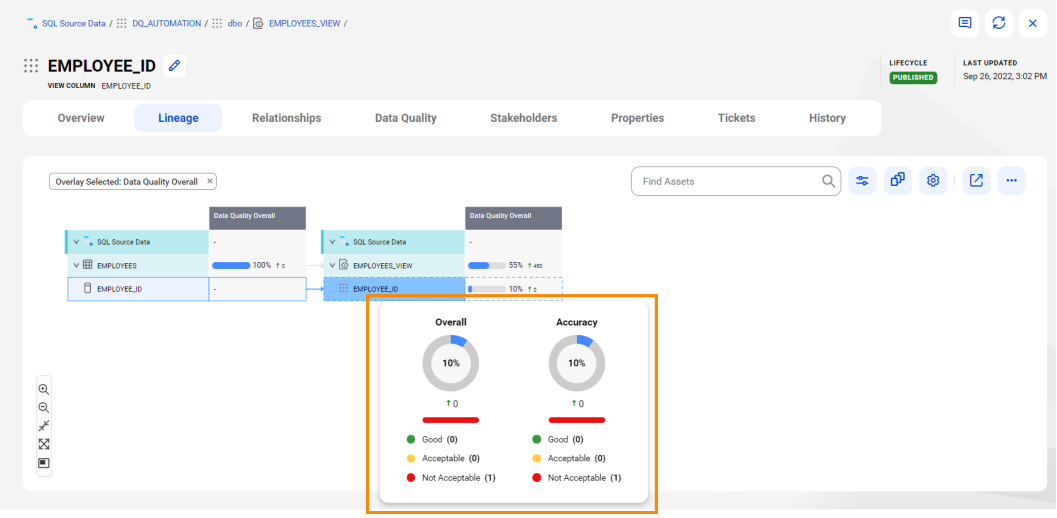
To see the details of rule occurrences that constitute the data quality score, go to the **Data Quality** tab of the asset. The following image shows the data quality rule occurrences on the **Data Quality** tab of a business term:



Lineage tab of an asset

When you open an asset, the **Lineage** tab depicts the flow of data in the form of a visual representation from the system of its origin to the system of its destination in your organization. To view the data quality scores for the rule that is run on the data element, select one of the data quality options from the **Overlays** options.

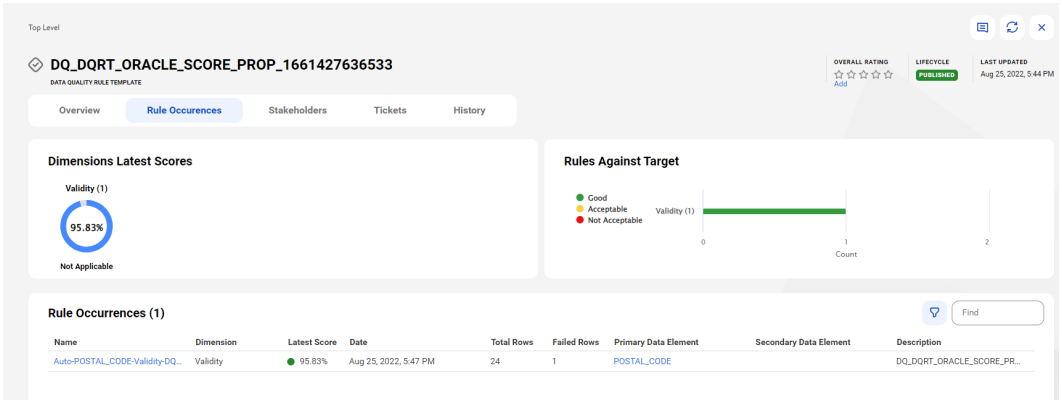
The following image shows the data quality score in the **Lineage** tab of a data element:



Rule Occurrences tab of a data quality rule template

When you open a rule template, the **Rule Occurrences** tab shows all rule occurrences that are manually created and automatically generated from the rule template. A rule template is the governance representation of a data quality rule that you want to run on some data elements. The rule occurrences are the individual scores of data quality rules that are run on each data element.

The following image shows the **Rule Occurrences** tab of a data quality rule template:



Viewing latest preview of failed rows

You can view the latest data preview of failed rows for a particular rule occurrence in the **Rule Occurrences** tab of data quality rule template. By default, the system can retrieve the data from a maximum of 100 failed rows.

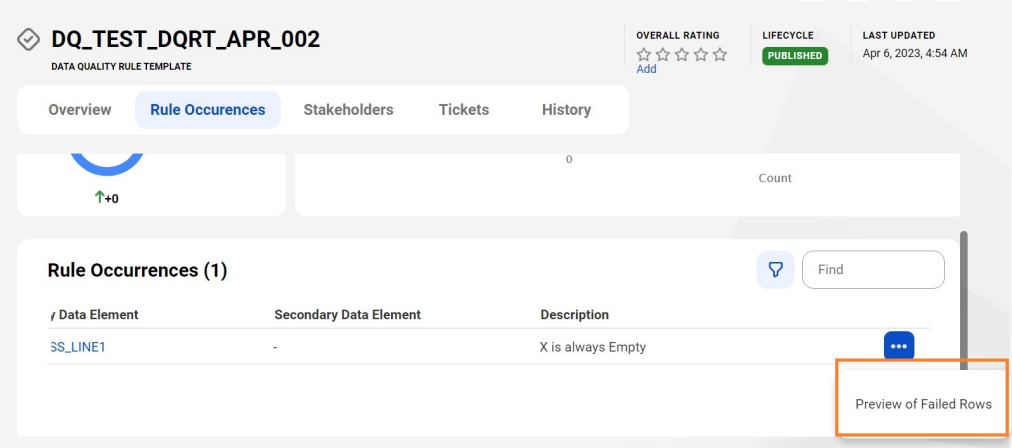
To view the data preview of failed rows, the following prerequisites apply:

- Ensure that the organization administrator has enabled the **Preview Data** feature privilege for your user role in Administrator.

- To perform drill down, and to view the **Preview of Successful Rows** and **Preview of Unsuccessful Rows** in a Data Governance and Catalog scorecard, you need the following privileges for the Data Profiling service in Administrator:

  - **Drill Down**
  - **Query - Create**
  - **Query - Submit**
  - **Data Profiling Results - View**
  - **Sensitive Data - View**

  For more information about the privileges in Data Profiling, see Data Profiling feature privileges in the *Administrator* help.
- The data quality rule is run at least once, and data quality scores are available to retrieve the data preview of failed rows.
- The primary data element is associated with the rule occurrence and is not of a hierarchical type.
- The rule occurrence is associated with a rule template.
- Data Governance and Catalog is integrated with a data quality application and the **Data Quality** option is enabled in Metadata Command Center for the catalog source.



In the **Rule Occurrences** tab of data quality rule template, click the Action menu for the rule occurrence for which you want to view the failed rows and select the **Preview of Failed Rows** option. The failed rows for a particular rule occurrence are displayed that are generated from the rule template.

Note: In Metadata Command Center, when you run a catalog source with the **Agent Cache** option enabled in the **Cache Result** field in the **Data Quality** capability, and then run a data quality job in Data Governance and Catalog, the preview of the failed rows is automatically generated. Thus, when you click the **Preview of Failed Rows** option, you can immediately view the failed rows table. However, when you run the catalog source with the **No Cache** option enabled in the **Cache Result** field, a preview job is triggered in Metadata Command Center. As a result, the preview of the failed rows takes a longer time to be generated.

Score tab of a data quality rule occurrence

When you open a rule occurrence, the **Score** tab shows the exact data quality score for the rule that is run on a specific data element. The tab also shows the acceptance criteria of the score according to the threshold

and target values you have specified in the rule template. Additionally, the tab shows the number of times the rule was run for the data element in the data quality application.

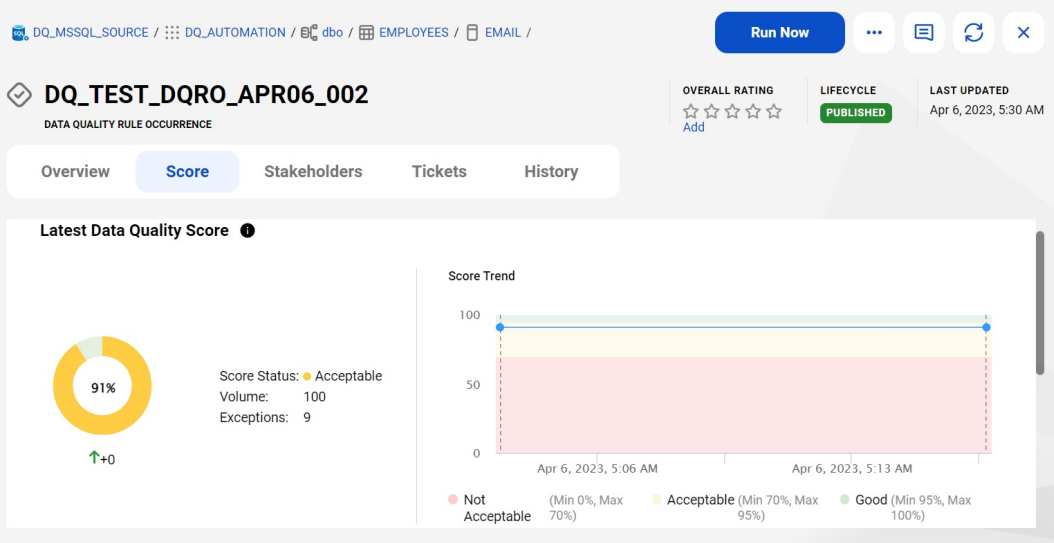
Note: The data quality scores are calculated only when the workflow for a data quality rule occurrence is in **Published** lifecycle state. If the workflow for a data quality rule occurrence is in **Draft** or **In-review** state, the scores are not calculated.

Before you run a data quality rule manually, the following prerequisites apply:

- The **Data Quality** option for the catalog source is enabled in Metadata Command Center.
- Your organization administrator has granted you the **Execute** permission on data quality rule occurrence assets through access policies in Metadata Command Center.
- You're a stakeholder of the data quality rule occurrence.
- The data quality rule occurrence is in the **Published** lifecycle status.

To run a data quality rule manually, go to the **Score** tab of the rule occurrence. Click the **Run Now** option in the top right corner of the page to trigger a job that retrieves the latest data quality score for the rule occurrence. When the job is triggered successfully, you can monitor the job in Metadata Command Center.

The following image shows the **Score** tab of a data quality rule occurrences:



Viewing latest preview of failed rows

You can view the latest data preview of failed rows for a particular rule occurrence in the **Score** tab of data quality rule occurrence. By default, the system can retrieve the data from a maximum of 100 failed rows.

To view the data preview of failed rows, the following prerequisites apply:

- Ensure sure that the organization administrator grants the **Preview Data** feature privilege for your user role in Administrator.
- The data quality rule is run at least once, and data quality scores are available to retrieve the data preview of failed rows.
- The primary data element is associated with the rule occurrence and is not of a hierarchical type.
- The rule occurrence is associated with a rule template.
- Data Governance and Catalog is integrated with a data quality application and the **Data Quality** option is enabled in Metadata Command Center for a source system.

In the **Score** tab of data quality rule occurrence, select the **Preview of Failed Rows** option from the **Data Preview** drop-down in the **Total Runs** section. The failed rows are displayed for the particular rule occurrence.

Note: In Metadata Command Center, when you run a catalog source with the **Agent Cache** option enabled in the **Cache Result** field in the **Data Quality** capability, and then run a data quality job in Data Governance and Catalog, the preview of the failed rows is automatically generated. Thus, when you click the **Preview of Failed Rows** option, you can immediately view the failed rows table. However, when you run the catalog source with the **No Cache** option enabled in the **Cache Result** field, a preview job is triggered in Metadata Command Center. As a result, the preview of the failed rows takes up to one hour to be generated.

Note: When you upload data quality scores in Data Governance and Catalog using bulk upload, the scores are automatically rounded to two decimal places. If you set a score with more than two decimal points, the score is rounded to the nearest value and retains only two decimal places.

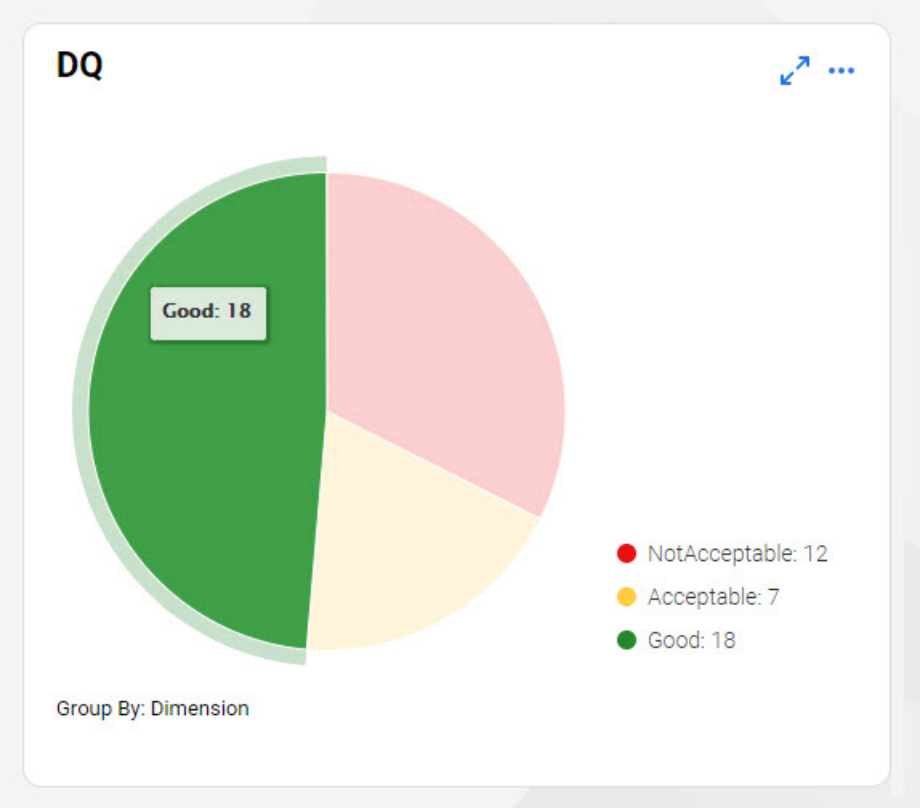
Data Quality widget

On the dashboard, you can add a widget that displays the cumulative data quality result of all rule runs.

You can specify the widget to show the data quality result in several ways. For more information about creating a data quality widget, see the *Asset Discovery* help.

Data quality result as per acceptance scores

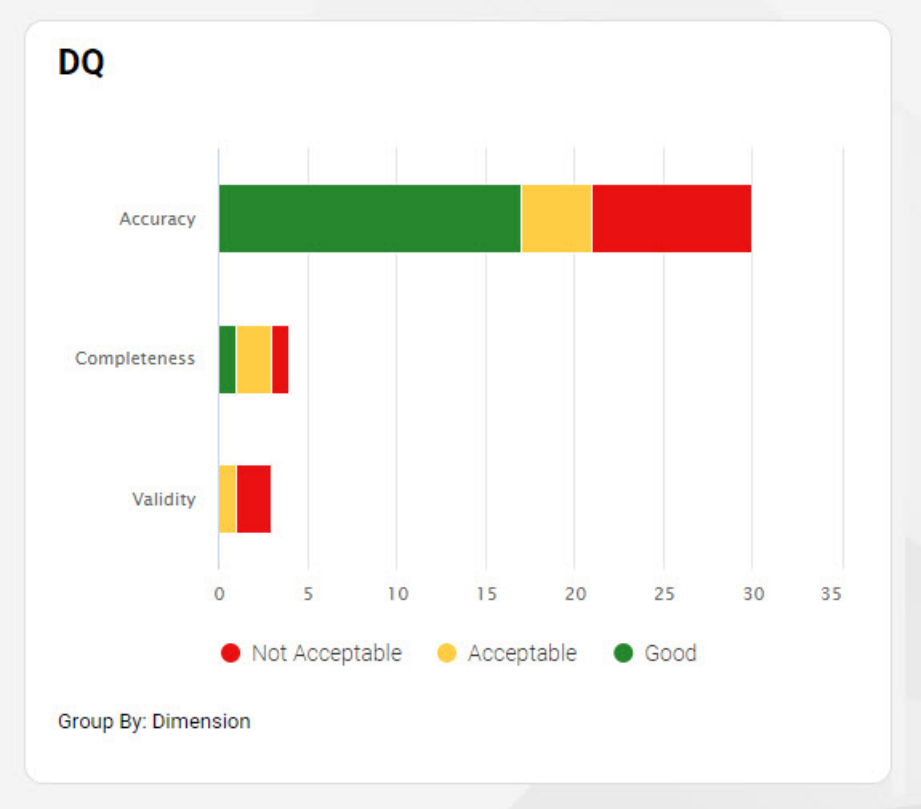
View a pie chart that shows the number of rule occurrences for each acceptance score. The following image shows a pie chart data quality widget:



Click an acceptance score color to see the list of rule occurrences that contribute to the score.

Data quality result for each dimension

View a bar chart to see the number of rule occurrences for each acceptance score according to the data quality dimensions. The following image shows a bar chart data quality widget:

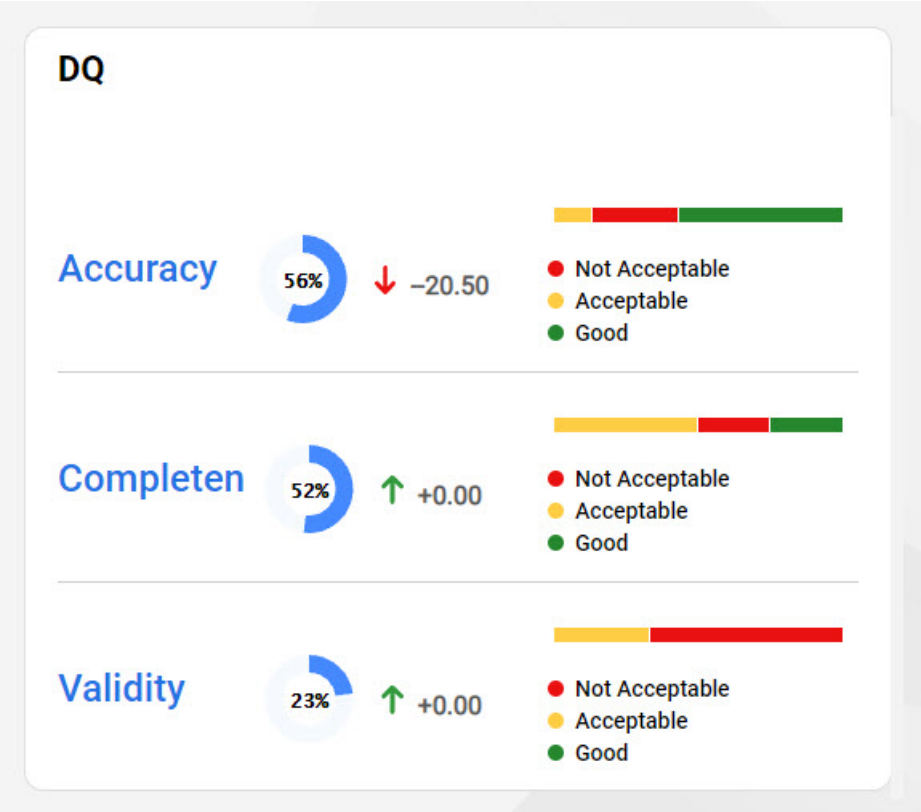


.

Click an acceptance score color to see the list of rule occurrences that contribute to the score for a particular dimension.

Composite score for each dimension

View a score chart to see the composite score for the rule occurrences according to the data quality dimensions. The following image shows a score chart data quality widget:



In the widget, you can see the following information:

- Composite score for each dimension.
- Increase or decrease in score for each dimension since the last rule run.
- Breakdown of the rule occurrences for each dimension according to the acceptance score.

For more information about composite score, see "How composite scores are calculated" on page 46.

CHAPTER 4

View notifications for data quality scores

When there is a change in the data quality score for a data quality rule occurrence, Data Governance and Catalog generates a notification and an email alert. You will receive notifications for a data quality rule occurrence status change only if you are assigned as a stakeholder for that rule occurrence.

You can receive notifications for the following changes to the rule occurrences:

- The status changes from good to not acceptable
- The status changes from acceptable to not acceptable
- The status changes from good to acceptable
- The status changes from not acceptable to good
- The status changes from not acceptable to acceptable
- The status changes from acceptable to good

You'll also receive notifications when the data quality score status is not acceptable after the first catalog source run.

Enabling email notifications

You can configure email notifications in Data Governance and Catalog to receive emails that will notify you when the data quality score of a data quality rule occurrence drops.

To configure email notifications, perform the following steps:

1. On the **Notification** menu, click the **Settings** icon.
2. On the **Notification Settings** panel, enable the *Email Event* option for the **Data Quality** notification type.

CHAPTER 5

Interpret data quality scores

Data Governance and Catalog displays data quality scores for assets in various graphical formats. Depending on the asset type that you open, the scores can be a composite score, the individual scores that make up the composite score, or details of specific rule runs on individual data elements.

Composite score for an asset

When you open an asset, the **Overview** tab displays the composite score for the asset on the **Data Quality** panel. The score represents the data quality result for all constituent data elements across all data quality dimensions, and appears as a donut chart.

The following image shows the components of a composite score:



1. Composite data quality score for the asset. The composite score is the average value of all data elements that are represented by the asset.
2. Increase or decrease in score since the last rule run.
3. Date of the last rule run. The date indicates the recency and freshness of the score.

To understand the logic behind a composite score, see "How composite scores are calculated" on page 46.

Asset types that display composite scores

You can view composite scores for the following types of assets:

- Business Term. Data elements that are linked to Glossary Business Term assets.
- System. Data sets that are linked to System assets.
- Data Set. Data elements that are linked to Data Set assets.
- Technical assets. Supported technical assets and their data elements. For more information about supported technical assets, see "Supported assets for automated data quality rules" on page 7.
- Rule Template. Data elements that are linked to Glossary assets that you specify in a rule template.
- Policy. Data sets and data elements that are linked to Policy assets.
- Process. Data elements that are linked to Process assets.

How composite scores are calculated

The composite score for an asset is derived by calculating the simple average of the rule occurrence scores of immediate child assets or directly related assets.

When you open a **technical asset**, the composite score is the simple average value of rule occurrence scores for all constituent data elements that are immediate child assets. If you open a data element, the composite score is the rule occurrence score for the data element. When you open a **business asset**, the composite score is the simple average value of the rule occurrence scores for all directly related assets.

Example

Consider a table called 'Oct2021Revenue' with rule occurrence scores for four data elements:

Week1Revenue	94%
Week2Revenue	86%
Week3Revenue	78%
Week4Revenue	82%

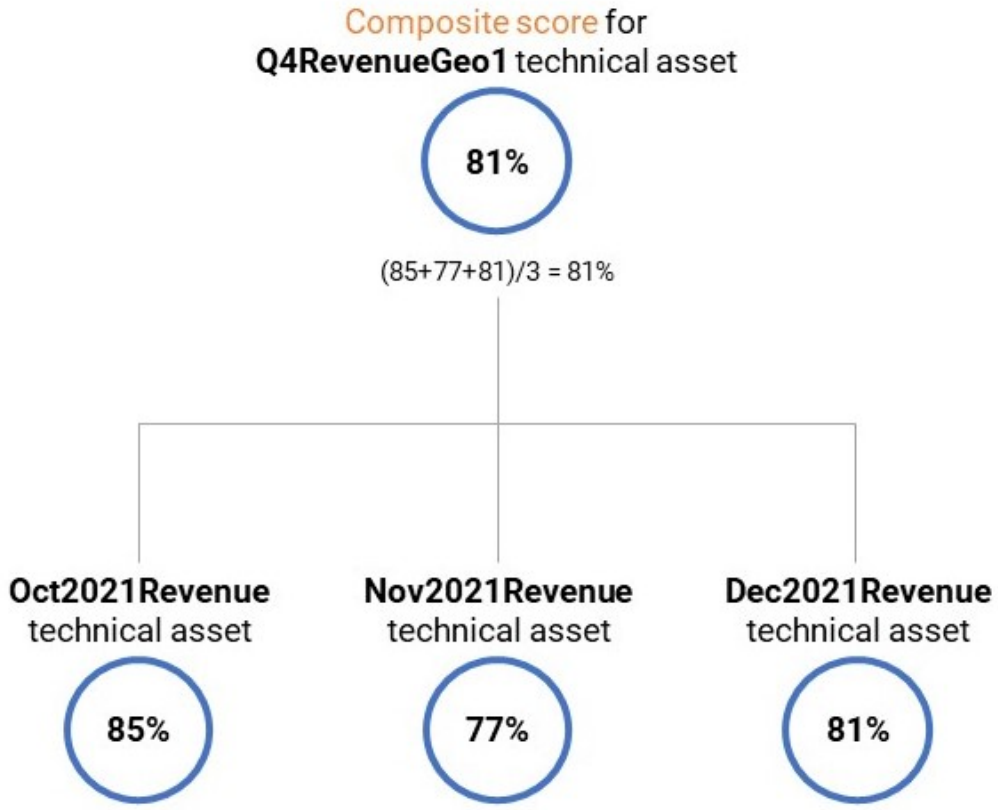
The composite score for the 'Oct2021Revenue' technical asset is the average value of the four rows. Therefore, In this example, the composite score for the 'Oct2021Revenue' technical asset is (94+86+78+82)/4 = 85%. The following diagram depicts this composite score:



Similarly, we can have the following composite scores:

- Composite score for the 'Oct2021Revenue' technical asset = 85%
- Composite score for the 'Nov2021Revenue' technical asset = 77%
- Composite score for the 'Dec2021Revenue' technical asset = 81%

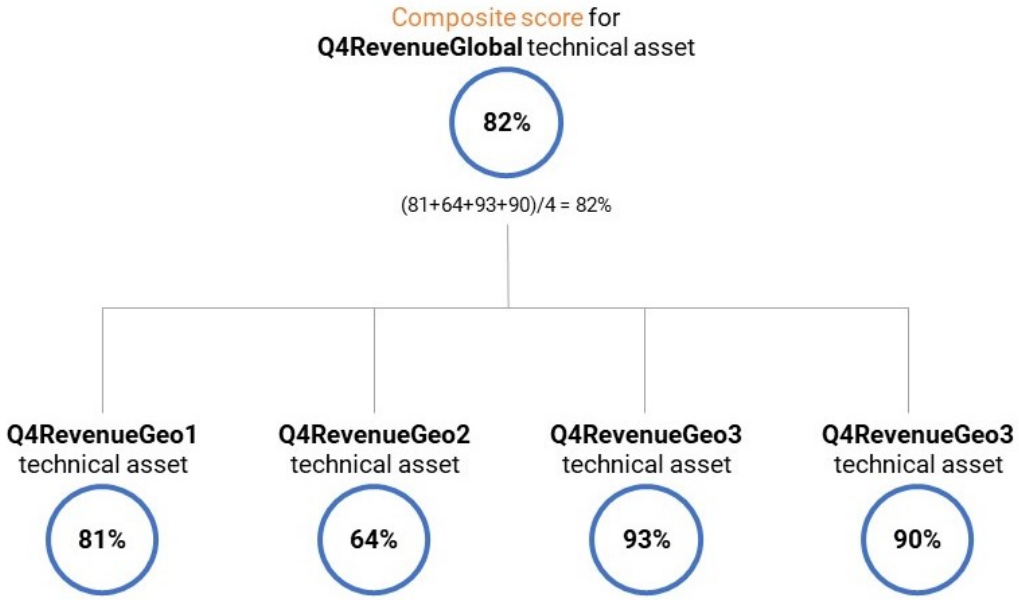
Now, if we have a data container called 'Q4RevenueGeo1' that contains the above three technical assets, the composite score for the 'Q4Revenue' technical asset is (85+77+81)/3 = 81%. The following diagram depicts this composite score:



Similarly, we can have the following composite scores:

- Composite score for the 'Q4RevenueGeo1' technical asset = 81%
- Composite score for the 'Q4RevenueGeo2' technical asset = 64%
- Composite score for the 'Q4RevenueGeo3' technical asset = 93%
- Composite score for the 'Q4RevenueGeo4' technical asset = 90%

Now, if we have a data container called 'Q4RevenueGlobal' that contains the above four technical assets, the composite score for the 'Q4Revenue' technical asset is (81+64+93+90)/4 = 82%. The following diagram depicts this composite score:

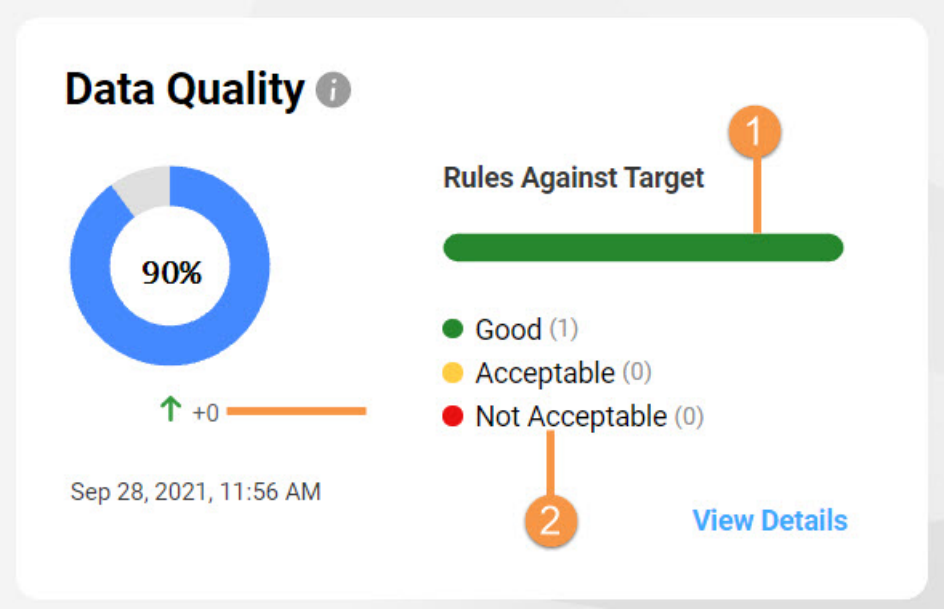


In each case, the composite score is derived by calculating the average rule scores of the immediate child assets. The composite score does not include rule scores of subsequent child assets beyond the first level.

Acceptance score for an asset

When you open an asset, the **Data Quality** panel on the **Overview** tab additionally displays the acceptance score for the asset. The score appears as a horizontal bar.

The following image shows the components of an acceptance score:

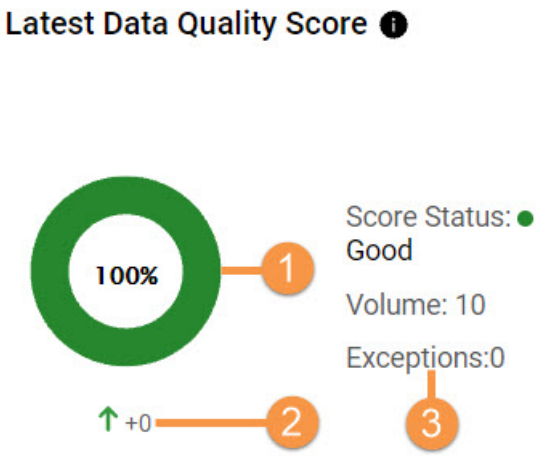


1. Rules run on data elements that meet the defined criteria of Good, Acceptable, and Not Acceptable. You can define the criteria in the rule template as follows:
   - Good: A score above the target value, represented in green color.
   - Acceptable: A score between the threshold and target value, represented in amber color.
   - Not Acceptable: A score below the threshold value, represented in red color.
2. Number of rule occurrences for each acceptance score.

Individual score for a data element

When you open a rule occurrence, the **Score** tab displays the data quality score for the rule that is run on a specific data element. The score is the composite score for the data element across all data quality dimensions. The score appears as a donut chart.

The following image shows the components of an individual score:



1. Individual data quality score for the asset across all dimensions.
2. Increase or decrease in score since the last rule run.
3. Additional details about the score, as follows:

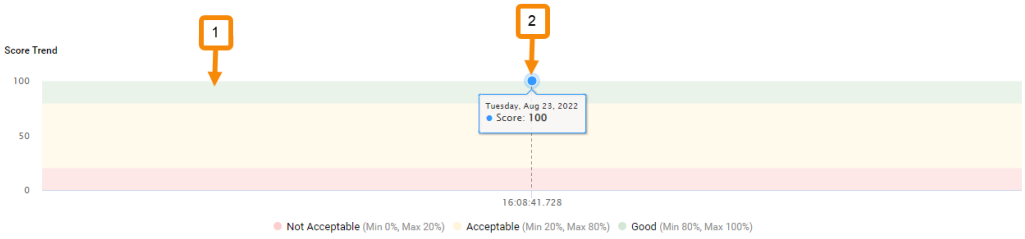
Label	Description
Score Status	Acceptance score for the asset according to the target and threshold values defined in the rule template.
Volume	Total number of rows on which the data quality rule was run.
Exceptions	Total number of rows in the data element for which the data quality rule failed.

To understand how the scores are calculated, see "How composite scores are calculated" on page 46.

Score trend for a data element

When you open a rule occurrence, the **Score** tab additionally displays the trend of the data quality score over several runs. The trend appears as a horizontal bar chart that depicts the acceptance values defined in the rule template.

The following image shows the components of a score trend:



1. Target and threshold values of the rule that define the acceptance score. In this image, the threshold value is 50 and the target value is 70.
2. Hover your mouse over a blue dot to see the date of a rule run and the rule occurrence that was created during the rule run.

CHAPTER 6

Data observability

Use Data Governance and Catalog to visualize and evaluate the events that data observability generates for anomalies identified in your source data. Data observability identifies the anomalies through the analysis of sudden deviations from a usual trend observed within your source system.

For the anomalies that it identifies, data observability generates events in Data Governance and Catalog. Data observability categorizes the events based on metric and severity. You can take appropriate actions on the anomaly events. Data observability learns your actions so that it can generate better events in the future.

To receive event notifications, the administrator must enable data observability for a source system in Metadata Command Center.

View data freshness and data volume

Monitor the time intervals at which your source data is updated and refreshed, along with volume metrics.

Data Freshness

The freshness data allows you to observe how often and in what pattern your source data is updated real-time. You can view the freshness data in the **Freshness** tab of the **Freshness and Volume** panel in Data Governance and Catalog.

You can track both freshness and volume for the Snowflake catalog source.

Data Volume

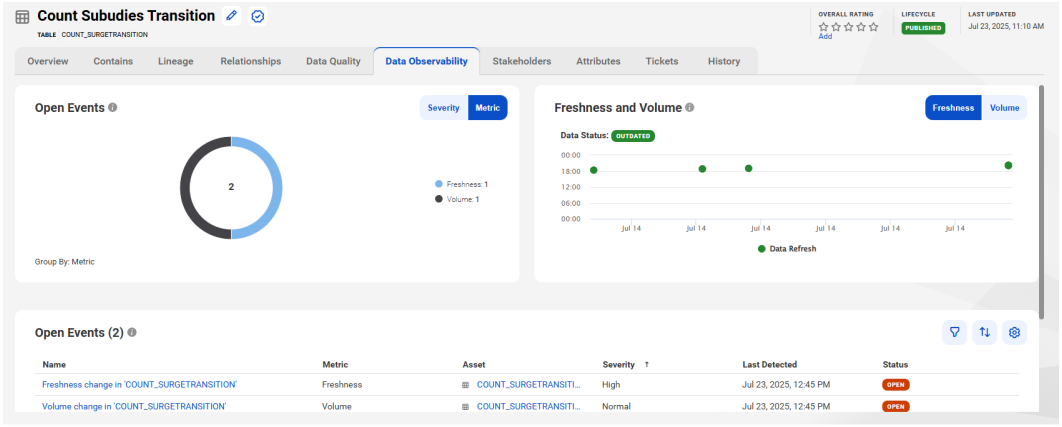
The volume data allows you to observe in real time the row count of your source data that are updated and refreshed. You can track the total number of rows in your data and the changes in row count each time your data is refreshed.

You can track the volume metric for the following catalog sources:

- Oracle
- MariaDB
- Google BigQuery
- Microsoft Fabric Data Lakehouse
- Microsoft Fabric Data Warehouse
- Microsoft SQL
- Microsoft Azure Synapse
- MySQL

- SAP S/4HANA
- Snowflake

The following image shows the data freshness and data volume panel in a data set:



You can generate event anomalies of your data freshness and volume by selecting **Data Freshness** and **Data Volume** filter options in Metadata Command Center. For more information about anomaly events, see "View anomaly events" on page 53.

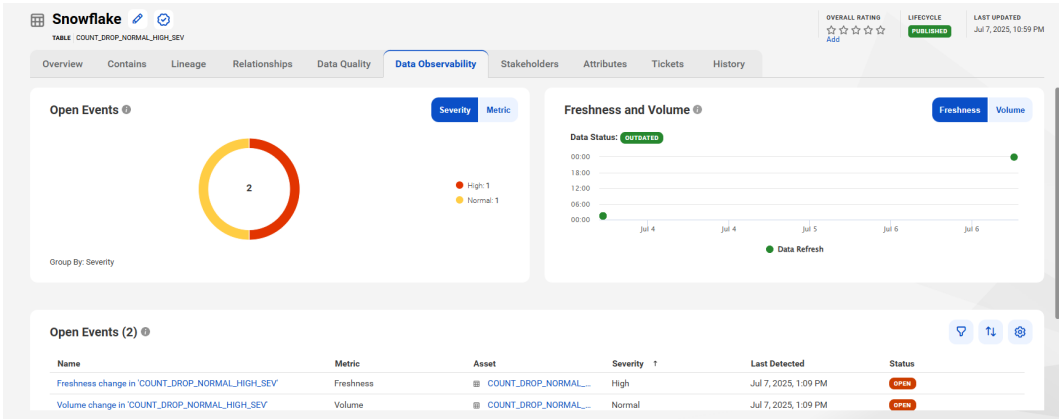
View anomaly events

View the events that data observability generates when anomalies are detected for technical assets. Among technical assets, you can examine anomaly events generated for catalog sources, technical data sets, and data elements.

To access the events generated for a specific asset, browse or search for the technical asset for which you want to see the events that data observability generated. Click the asset to open the asset details page, and go to the **Data Observability** tab to view the anomaly events generated for the asset.

To access the freshness and volume data of a catalog source, click the table data set of the catalog source and go to the **Data Observability** tab to view the anomaly events generated for the catalog source in the **Metric** tab of the **Open Events** panel.

The following image shows the open events for a column asset:



On the **Data Observability** tab, analyze the events generated for the data observability anomalies identified in your source data. To review the individual events generated when anomalies are identified for a specific asset, select each event listed on the **Open Events** panel.

You can now specify the maximum number of anomaly events to generate for each catalog source run in Metadata Command Center. You can enter any value between 10 and 20000.

Note: The following guidelines apply to data observability anomaly events:

- The administrator of the catalog source might have applied filters to the data in Metadata Command Center to narrow down the data elements that are applicable for business users. The data for which you receive anomaly notifications depend either on the metadata extraction filters or profiling filters that the administrator has configured for the catalog source.
- If the detected anomalies in a catalog source exceed the limit defined in the **Maximum Events to Generate** field in Metadata Command Center, you see a single event instead of individual events. The single event shows aggregated information on the detected anomalies.
- If an anomaly is detected in the catalog source and if you are a stakeholder of the catalog source, you can choose to receive email notifications for the anomalies. The email summarizes the data observability task job that was run.
- If no anomalies are detected, or anomalies are detected in a failed, skipped, or a partially-complete data observability job, you will not receive any email notification.

To understand the various types of anomalies and their corresponding metrics in data observability, see the How-to Library article Understand Data Observability Anomalies in Data Governance and Catalog.

Event operations

When you analyze the events that data observability generates for the anomalies observed in the characteristics of the data in your source system, you can resolve and reject the events that data observability generates. Data observability learns from your response so that it can provide more accurate events in the future.

You can perform the following operations for the events:

Resolve an event

If you are a stakeholder of a technical asset where the anomalies were observed, and if you find that the anomaly event is a correct observation, you can mark the event as valid. If a similar anomaly is detected again, data observability will create a new event to indicate the anomaly. To mark the event as valid, click **Resolve** on the event page.

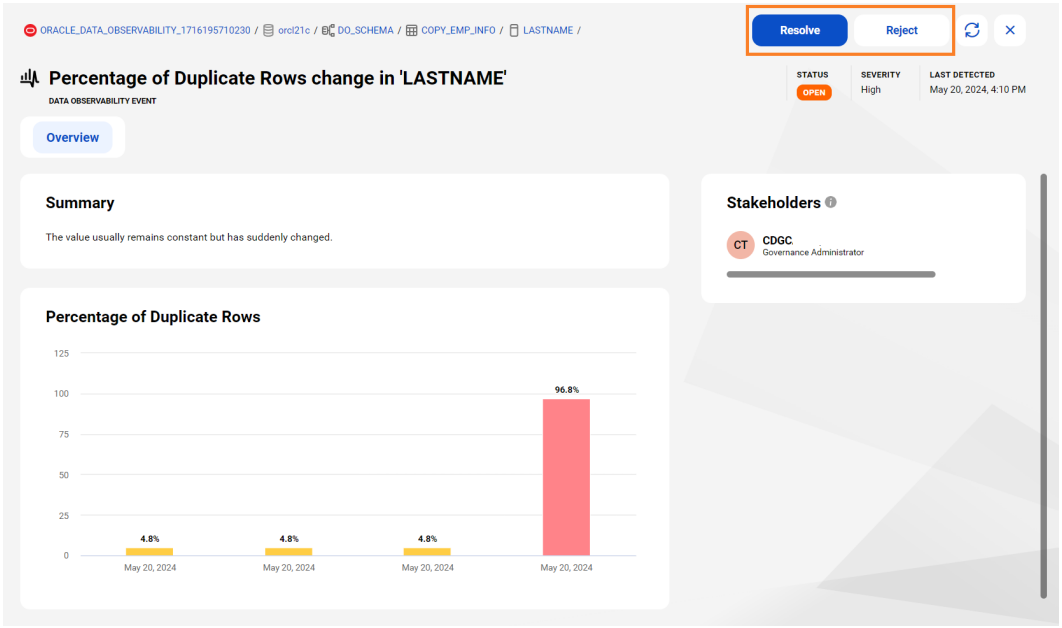
For example, if data observability generates an event due to a sudden increase in the number of rows added to your source data from 10 to 10,000, verify whether the addition of 10,000 rows constitutes a valid anomaly that data observability identified. If you consider the anomaly identified by data observability to be valid, you can resolve it to address the generated event.

Reject an event

If you are a stakeholder of a technical asset where the anomalies were observed, and if you find that the anomaly event is an incorrect observation because you expected the change in your source data, you can mark the event as invalid. If a similar anomaly is detected again, data observability will not notify you. To mark the event as invalid, click **Reject** on the event page.

For example, if data observability generates an event due to a sudden increase in the number of rows added to your source data from 10 to 10,000, verify whether the addition of 10,000 rows constitutes an expected change. If you determine that the anomaly identified by data observability is an expected change, you can reject it to address the generated event.

The following image shows the **Resolve** and **Reject** operations for an event:



On an event page, you will learn about the type of anomaly that data observability has observed. Additionally, you will see a trend chart for percentage changes in metric values for the anomalies detected in your source data for the catalog source jobs that you run. The page also displays the stakeholders who can perform resolve and reject operations for events.

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